Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 12:21:11

Measurement Details:

- **Sample:** Muestra7
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

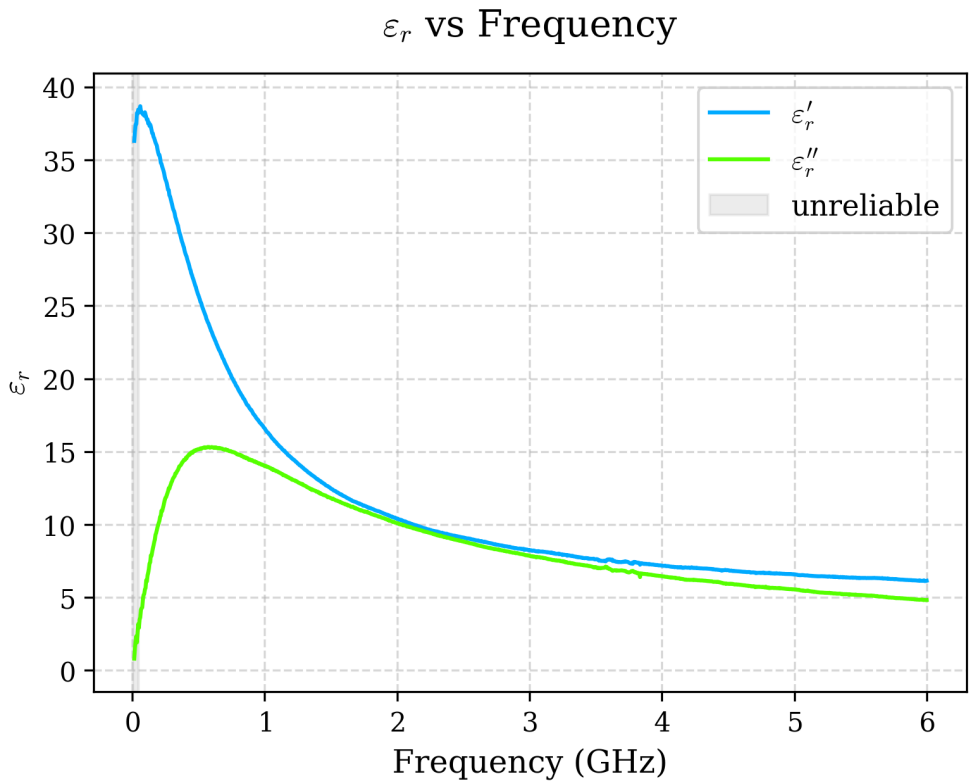
Contents

1 Measurement Results **2**
1.1 Complex Permittivity $\varepsilon_r(f)$ 2
1.2 S_{11} — Smith Chart 29

2 Measurement Quality **56**

1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	36.3039	0.8098	0.0223
0.020	36.9800	1.5735	0.0425
0.025	37.4209	2.0021	0.0535
0.030	37.4695	2.3463	0.0626
0.035	38.2143	1.9164	0.0501
0.040	38.1706	2.6089	0.0683
0.045	38.4501	3.1868	0.0829
0.050	38.4565	2.9228	0.0760
0.055	38.2714	3.4905	0.0912
0.060	38.6810	3.6390	0.0941
0.065	38.3113	4.1526	0.1084
0.070	38.2775	4.2710	0.1116
0.075	38.2000	4.3177	0.1130
0.080	38.1234	4.8906	0.1283
0.085	38.1642	5.1688	0.1354

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	38.0277	5.2341	0.1376
0.095	38.2645	5.5537	0.1451
0.100	38.0865	5.6530	0.1484
0.105	37.9204	6.1839	0.1631
0.110	37.7550	6.1029	0.1616
0.115	37.7802	6.5116	0.1724
0.120	37.4940	6.7005	0.1787
0.125	37.5669	6.9616	0.1853
0.130	37.3268	7.3339	0.1965
0.135	37.4232	7.4143	0.1981
0.140	37.2803	7.8895	0.2116
0.145	37.0527	7.8589	0.2121
0.150	36.8339	8.1252	0.2206
0.155	36.8104	8.3534	0.2269
0.160	36.5844	8.5562	0.2339
0.165	36.4967	8.7847	0.2407
0.170	36.3516	8.9688	0.2467
0.175	36.1951	9.2962	0.2568
0.180	36.0907	9.4223	0.2611
0.185	35.9800	9.6173	0.2673
0.190	35.7215	9.8160	0.2748
0.195	35.5100	9.9787	0.2810
0.200	35.3421	10.1254	0.2865
0.205	35.3345	10.2712	0.2907
0.210	35.1143	10.5333	0.3000
0.215	34.7961	10.6071	0.3048
0.220	34.7313	10.9509	0.3153
0.225	34.5522	10.9157	0.3159
0.230	34.4097	11.0867	0.3222
0.235	34.2699	11.2641	0.3287
0.240	34.0855	11.4880	0.3370
0.245	33.8429	11.6793	0.3451
0.250	33.6618	11.8289	0.3514
0.255	33.5547	11.8792	0.3540
0.260	33.4049	12.0148	0.3597
0.265	33.1750	12.1601	0.3665
0.270	32.9479	12.2847	0.3729
0.275	32.9329	12.4922	0.3793
0.280	32.5687	12.4634	0.3827
0.285	32.4981	12.6276	0.3886
0.290	32.3070	12.7390	0.3943
0.295	32.1274	12.8911	0.4012
0.300	31.9552	12.9690	0.4058
0.305	31.7400	13.1122	0.4131
0.310	31.5819	13.1981	0.4179
0.315	31.4958	13.2596	0.4210

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	31.3286	13.3279	0.4254
0.325	31.0748	13.4998	0.4344
0.330	30.9375	13.5068	0.4366
0.335	30.7646	13.6083	0.4423
0.340	30.6187	13.6854	0.4470
0.345	30.4898	13.7500	0.4510
0.350	30.2522	13.8743	0.4586
0.355	30.0451	13.9919	0.4657
0.360	29.9234	14.0251	0.4687
0.365	29.7523	14.0711	0.4729
0.370	29.5745	14.1298	0.4778
0.375	29.4016	14.2456	0.4845
0.380	29.2794	14.2797	0.4877
0.385	29.1169	14.3585	0.4931
0.390	28.9942	14.4334	0.4978
0.395	28.7654	14.4213	0.5013
0.400	28.6042	14.4504	0.5052
0.405	28.4732	14.5825	0.5121
0.410	28.3743	14.5550	0.5130
0.415	28.1862	14.6606	0.5201
0.420	28.0324	14.7105	0.5248
0.425	27.9019	14.7472	0.5285
0.430	27.7359	14.8368	0.5349
0.435	27.5326	14.8008	0.5376
0.440	27.4300	14.8675	0.5420
0.445	27.2214	14.9480	0.5491
0.450	27.1144	14.9572	0.5516
0.455	26.9529	14.9191	0.5535
0.460	26.8299	15.0180	0.5597
0.465	26.6913	15.0197	0.5627
0.470	26.5350	15.0760	0.5682
0.475	26.3885	15.0638	0.5708
0.480	26.2786	15.1148	0.5752
0.485	26.0975	15.1104	0.5790
0.490	25.9608	15.1431	0.5833
0.495	25.8083	15.1604	0.5874
0.500	25.6987	15.1511	0.5896
0.505	25.5601	15.1802	0.5939
0.510	25.4253	15.1882	0.5974
0.515	25.2591	15.2259	0.6028
0.520	25.1923	15.1854	0.6028
0.525	25.0263	15.2291	0.6085
0.530	24.8794	15.2517	0.6130
0.535	24.7624	15.2629	0.6164
0.540	24.6352	15.2795	0.6202
0.545	24.5104	15.2974	0.6241

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	24.3952	15.2602	0.6255
0.555	24.2764	15.2935	0.6300
0.560	24.0949	15.2768	0.6340
0.565	23.9975	15.2843	0.6369
0.570	23.8693	15.3188	0.6418
0.575	23.7967	15.3154	0.6436
0.580	23.6683	15.3152	0.6471
0.585	23.5447	15.2707	0.6486
0.590	23.4271	15.2761	0.6521
0.595	23.2861	15.3069	0.6573
0.600	23.2050	15.3149	0.6600
0.605	23.0688	15.2959	0.6631
0.610	22.9426	15.2778	0.6659
0.615	22.8464	15.2942	0.6694
0.620	22.7482	15.2563	0.6707
0.625	22.6231	15.2829	0.6755
0.630	22.5451	15.2615	0.6769
0.635	22.3995	15.2830	0.6823
0.640	22.3118	15.2558	0.6838
0.645	22.1865	15.2237	0.6862
0.650	22.0768	15.2467	0.6906
0.655	21.9731	15.2572	0.6944
0.660	21.9042	15.2163	0.6947
0.665	21.8104	15.2268	0.6981
0.670	21.6798	15.2241	0.7022
0.675	21.5991	15.1971	0.7036
0.680	21.4616	15.1975	0.7081
0.685	21.3773	15.1658	0.7094
0.690	21.3085	15.1481	0.7109
0.695	21.1814	15.1299	0.7143
0.700	21.0712	15.1185	0.7175
0.705	20.9842	15.1622	0.7226
0.710	20.8881	15.1083	0.7233
0.715	20.7868	15.0977	0.7263
0.720	20.6856	15.1135	0.7306
0.725	20.6012	15.0711	0.7316
0.730	20.5214	15.0478	0.7333
0.735	20.3881	15.0296	0.7372
0.740	20.3257	15.0378	0.7398
0.745	20.2312	15.0284	0.7428
0.750	20.0943	14.9950	0.7462
0.755	20.0365	14.9738	0.7473
0.760	19.9444	14.9764	0.7509
0.765	19.8565	14.9598	0.7534
0.770	19.7691	14.9509	0.7563
0.775	19.7225	14.9170	0.7563

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	19.5958	14.9082	0.7608
0.785	19.5061	14.8674	0.7622
0.790	19.4606	14.8180	0.7614
0.795	19.3631	14.8248	0.7656
0.800	19.2726	14.7879	0.7673
0.805	19.1779	14.7918	0.7713
0.810	19.1063	14.7661	0.7728
0.815	19.0319	14.7856	0.7769
0.820	18.9514	14.7503	0.7783
0.825	18.8682	14.6931	0.7787
0.830	18.8160	14.7130	0.7819
0.835	18.7175	14.6692	0.7837
0.840	18.6323	14.6171	0.7845
0.845	18.5993	14.6264	0.7864
0.850	18.5061	14.5928	0.7885
0.855	18.4096	14.6120	0.7937
0.860	18.3306	14.5425	0.7933
0.865	18.2267	14.5653	0.7991
0.870	18.1759	14.5537	0.8007
0.875	18.1576	14.5122	0.7992
0.880	18.0657	14.4404	0.7993
0.885	17.9801	14.4379	0.8030
0.890	17.9360	14.4623	0.8063
0.895	17.8544	14.4094	0.8071
0.900	17.8404	14.3980	0.8070
0.905	17.7493	14.4164	0.8122
0.910	17.6616	14.3801	0.8142
0.915	17.5694	14.3457	0.8165
0.920	17.5293	14.3131	0.8165
0.925	17.4659	14.3411	0.8211
0.930	17.4095	14.2994	0.8214
0.935	17.3188	14.2620	0.8235
0.940	17.2690	14.2740	0.8266
0.945	17.2034	14.2410	0.8278
0.950	17.1734	14.2032	0.8270
0.955	17.0953	14.2037	0.8309
0.960	17.0234	14.1636	0.8320
0.965	16.9704	14.1788	0.8355
0.970	16.9340	14.1403	0.8350
0.975	16.8663	14.1106	0.8366
0.980	16.8246	14.1157	0.8390
0.985	16.7861	14.0898	0.8394
0.990	16.6935	14.0639	0.8425
0.995	16.6439	14.0541	0.8444
1.000	16.5890	14.0071	0.8444
1.005	16.5183	14.0191	0.8487

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	16.4499	14.0191	0.8522
1.015	16.3738	13.9605	0.8526
1.020	16.3268	13.9591	0.8550
1.025	16.3042	13.9576	0.8561
1.030	16.2352	13.9183	0.8573
1.035	16.1853	13.9034	0.8590
1.040	16.1307	13.8881	0.8610
1.045	16.0622	13.8478	0.8621
1.050	15.9910	13.8450	0.8658
1.055	15.9295	13.7771	0.8649
1.060	15.8819	13.8086	0.8695
1.065	15.8443	13.7286	0.8665
1.070	15.7932	13.7499	0.8706
1.075	15.7165	13.7280	0.8735
1.080	15.6864	13.6949	0.8730
1.085	15.6493	13.6638	0.8731
1.090	15.5886	13.6685	0.8768
1.095	15.5192	13.6413	0.8790
1.100	15.4869	13.6259	0.8798
1.105	15.4372	13.5832	0.8799
1.110	15.3751	13.5618	0.8821
1.115	15.3406	13.5440	0.8829
1.120	15.3061	13.5243	0.8836
1.125	15.2630	13.4865	0.8836
1.130	15.1966	13.4728	0.8866
1.135	15.1355	13.4333	0.8875
1.140	15.1095	13.3786	0.8854
1.145	15.0573	13.4009	0.8900
1.150	15.0568	13.3685	0.8879
1.155	14.9738	13.3534	0.8918
1.160	14.9337	13.3100	0.8913
1.165	14.8893	13.3145	0.8942
1.170	14.8549	13.2690	0.8932
1.175	14.7754	13.2424	0.8963
1.180	14.7394	13.2414	0.8984
1.185	14.7210	13.2033	0.8969
1.190	14.6701	13.1658	0.8975
1.195	14.6316	13.1622	0.8996
1.200	14.5970	13.1452	0.9005
1.205	14.5332	13.0766	0.8998
1.210	14.5104	13.0840	0.9017
1.215	14.4593	13.0598	0.9032
1.220	14.4436	13.0445	0.9031
1.225	14.4049	13.0259	0.9043
1.230	14.3627	13.0200	0.9065
1.235	14.3003	12.9698	0.9070

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	14.2802	12.9699	0.9082
1.245	14.2370	12.9452	0.9093
1.250	14.1799	12.9050	0.9101
1.255	14.1424	12.8689	0.9100
1.260	14.1170	12.8576	0.9108
1.265	14.0544	12.8348	0.9132
1.270	14.0378	12.8088	0.9124
1.275	14.0056	12.8053	0.9143
1.280	13.9526	12.7686	0.9151
1.285	13.9243	12.7829	0.9180
1.290	13.8910	12.7102	0.9150
1.295	13.8330	12.6971	0.9179
1.300	13.7941	12.6748	0.9189
1.305	13.7718	12.6622	0.9194
1.310	13.7228	12.6315	0.9205
1.315	13.6955	12.6129	0.9210
1.320	13.6592	12.5624	0.9197
1.325	13.6078	12.5514	0.9224
1.330	13.5681	12.5212	0.9228
1.335	13.5377	12.4988	0.9233
1.340	13.5243	12.4837	0.9231
1.345	13.4734	12.4366	0.9230
1.350	13.4276	12.4379	0.9263
1.355	13.4083	12.3700	0.9226
1.360	13.3758	12.4036	0.9273
1.365	13.3356	12.3756	0.9280
1.370	13.3094	12.3543	0.9282
1.375	13.2830	12.3418	0.9291
1.380	13.2367	12.3201	0.9308
1.385	13.1947	12.2995	0.9322
1.390	13.1627	12.2573	0.9312
1.395	13.1191	12.2652	0.9349
1.400	13.1143	12.2533	0.9343
1.405	13.0578	12.1975	0.9341
1.410	13.0408	12.2076	0.9361
1.415	13.0040	12.1396	0.9335
1.420	12.9788	12.1549	0.9365
1.425	12.9580	12.0988	0.9337
1.430	12.8907	12.0904	0.9379
1.435	12.8813	12.0700	0.9370
1.440	12.8463	12.0426	0.9374
1.445	12.8135	12.0230	0.9383
1.450	12.7967	12.0233	0.9396
1.455	12.7515	11.9975	0.9409
1.460	12.7021	11.9900	0.9439
1.465	12.6769	11.9615	0.9436

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	12.6517	11.9266	0.9427
1.475	12.6393	11.8743	0.9395
1.480	12.6082	11.8955	0.9435
1.485	12.5695	11.8691	0.9443
1.490	12.5520	11.8525	0.9443
1.495	12.5088	11.8288	0.9456
1.500	12.4657	11.8093	0.9473
1.505	12.4407	11.7880	0.9475
1.510	12.4219	11.7485	0.9458
1.515	12.3882	11.7823	0.9511
1.520	12.3398	11.7406	0.9514
1.525	12.3202	11.7240	0.9516
1.530	12.3103	11.6952	0.9500
1.535	12.2921	11.6704	0.9494
1.540	12.2472	11.6511	0.9513
1.545	12.2071	11.6443	0.9539
1.550	12.2068	11.6254	0.9524
1.555	12.1773	11.6012	0.9527
1.560	12.1401	11.5862	0.9544
1.565	12.1063	11.5737	0.9560
1.570	12.0900	11.5333	0.9540
1.575	12.0547	11.5358	0.9570
1.580	12.0613	11.5024	0.9537
1.585	12.0225	11.4753	0.9545
1.590	11.9823	11.4629	0.9567
1.595	11.9613	11.4451	0.9568
1.600	11.9381	11.4268	0.9572
1.605	11.9206	11.4213	0.9581
1.610	11.9020	11.3763	0.9558
1.615	11.8652	11.3710	0.9583
1.620	11.8318	11.3770	0.9616
1.625	11.8146	11.3531	0.9609
1.630	11.7890	11.2981	0.9584
1.635	11.7664	11.3097	0.9612
1.640	11.7528	11.2969	0.9612
1.645	11.7107	11.2719	0.9625
1.650	11.6906	11.2207	0.9598
1.655	11.6755	11.2113	0.9602
1.660	11.6296	11.2108	0.9640
1.665	11.6125	11.2335	0.9674
1.670	11.5906	11.1901	0.9654
1.675	11.5737	11.1577	0.9641
1.680	11.5665	11.1634	0.9651
1.685	11.5585	11.1056	0.9608
1.690	11.5256	11.1147	0.9644
1.695	11.5188	11.0923	0.9630

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	11.4907	11.0706	0.9634
1.705	11.4594	11.0626	0.9654
1.710	11.4478	11.0274	0.9633
1.715	11.4529	11.0020	0.9606
1.720	11.3983	10.9913	0.9643
1.725	11.3951	10.9615	0.9619
1.730	11.3892	10.9801	0.9641
1.735	11.3560	10.9535	0.9646
1.740	11.3252	10.9390	0.9659
1.745	11.2903	10.9133	0.9666
1.750	11.3096	10.8912	0.9630
1.755	11.2870	10.9081	0.9664
1.760	11.2644	10.8772	0.9656
1.765	11.2548	10.8788	0.9666
1.770	11.2266	10.8556	0.9669
1.775	11.1919	10.8750	0.9717
1.780	11.1899	10.8227	0.9672
1.785	11.1425	10.8128	0.9704
1.790	11.1552	10.7854	0.9668
1.795	11.1387	10.7821	0.9680
1.800	11.1248	10.7462	0.9660
1.805	11.0883	10.7208	0.9669
1.810	11.0867	10.7158	0.9665
1.815	11.0673	10.6936	0.9662
1.820	11.0510	10.6818	0.9666
1.825	11.0293	10.6668	0.9671
1.830	10.9981	10.6398	0.9674
1.835	10.9784	10.6409	0.9693
1.840	10.9648	10.6227	0.9688
1.845	10.9539	10.6063	0.9683
1.850	10.9353	10.5883	0.9683
1.855	10.9193	10.5744	0.9684
1.860	10.9181	10.5422	0.9656
1.865	10.8985	10.5455	0.9676
1.870	10.8832	10.5324	0.9678
1.875	10.8495	10.5162	0.9693
1.880	10.8403	10.4920	0.9679
1.885	10.8113	10.4688	0.9683
1.890	10.8011	10.4559	0.9680
1.895	10.7776	10.4427	0.9689
1.900	10.7457	10.4350	0.9711
1.905	10.7426	10.4094	0.9690
1.910	10.7226	10.3887	0.9689
1.915	10.6902	10.3830	0.9713
1.920	10.6852	10.3580	0.9694
1.925	10.6634	10.3625	0.9718

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	10.6658	10.3147	0.9671
1.935	10.6365	10.3118	0.9695
1.940	10.6271	10.2616	0.9656
1.945	10.5817	10.2758	0.9711
1.950	10.5741	10.2511	0.9695
1.955	10.5493	10.2416	0.9708
1.960	10.5535	10.2180	0.9682
1.965	10.5197	10.1901	0.9687
1.970	10.5130	10.1808	0.9684
1.975	10.4792	10.1874	0.9722
1.980	10.4706	10.1455	0.9689
1.985	10.4742	10.1345	0.9676
1.990	10.4390	10.1236	0.9698
1.995	10.4241	10.1084	0.9697
2.000	10.3880	10.0958	0.9719
2.005	10.3977	10.0815	0.9696
2.010	10.3575	10.0756	0.9728
2.015	10.3526	10.0301	0.9689
2.020	10.3288	10.0229	0.9704
2.025	10.3128	10.0259	0.9722
2.030	10.3228	10.0055	0.9693
2.035	10.2908	10.0026	0.9720
2.040	10.2816	9.9886	0.9715
2.045	10.2518	9.9429	0.9699
2.050	10.2319	9.9756	0.9750
2.055	10.2275	9.9498	0.9728
2.060	10.1948	9.9344	0.9745
2.065	10.2048	9.9254	0.9726
2.070	10.1708	9.8890	0.9723
2.075	10.1513	9.8629	0.9716
2.080	10.1349	9.8727	0.9741
2.085	10.1196	9.8570	0.9741
2.090	10.1135	9.8505	0.9740
2.095	10.0955	9.8226	0.9730
2.100	10.0509	9.8249	0.9775
2.105	10.0560	9.8060	0.9751
2.110	10.0476	9.7975	0.9751
2.115	10.0197	9.7761	0.9757
2.120	10.0097	9.7468	0.9737
2.125	9.9842	9.7381	0.9754
2.130	9.9797	9.7274	0.9747
2.135	9.9666	9.7217	0.9754
2.140	9.9540	9.7106	0.9755
2.145	9.9321	9.6846	0.9751
2.150	9.9349	9.6833	0.9747
2.155	9.9057	9.6624	0.9754

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	9.8870	9.6494	0.9760
2.165	9.8822	9.6459	0.9761
2.170	9.8515	9.6228	0.9768
2.175	9.8455	9.6238	0.9775
2.180	9.8312	9.6054	0.9770
2.185	9.8150	9.5990	0.9780
2.190	9.7989	9.5814	0.9778
2.195	9.7681	9.5458	0.9772
2.200	9.7571	9.5387	0.9776
2.205	9.7589	9.5199	0.9755
2.210	9.7393	9.5278	0.9783
2.215	9.7274	9.5041	0.9770
2.220	9.7129	9.4944	0.9775
2.225	9.6869	9.4939	0.9801
2.230	9.6766	9.4721	0.9789
2.235	9.6650	9.4687	0.9797
2.240	9.6660	9.4422	0.9768
2.245	9.6300	9.4402	0.9803
2.250	9.6244	9.4197	0.9787
2.255	9.6241	9.4094	0.9777
2.260	9.5908	9.4023	0.9803
2.265	9.5793	9.3988	0.9812
2.270	9.5832	9.3579	0.9765
2.275	9.5555	9.3440	0.9779
2.280	9.5580	9.3325	0.9764
2.285	9.5405	9.3385	0.9788
2.290	9.5249	9.3133	0.9778
2.295	9.5359	9.2978	0.9750
2.300	9.5247	9.2682	0.9731
2.305	9.4875	9.2804	0.9782
2.310	9.4996	9.2436	0.9730
2.315	9.4924	9.2540	0.9749
2.320	9.4676	9.2403	0.9760
2.325	9.4561	9.2098	0.9740
2.330	9.4276	9.2478	0.9809
2.335	9.4293	9.2106	0.9768
2.340	9.4449	9.1889	0.9729
2.345	9.4066	9.1825	0.9762
2.350	9.3959	9.1695	0.9759
2.355	9.3971	9.1356	0.9722
2.360	9.3790	9.1539	0.9760
2.365	9.3675	9.1478	0.9765
2.370	9.3651	9.1130	0.9731
2.375	9.3710	9.0967	0.9707
2.380	9.3373	9.1100	0.9757
2.385	9.3199	9.0905	0.9754

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	9.3246	9.0883	0.9747
2.395	9.2918	9.0774	0.9769
2.400	9.2835	9.0595	0.9759
2.405	9.2899	9.0330	0.9723
2.410	9.2743	9.0352	0.9742
2.415	9.2638	9.0207	0.9738
2.420	9.2502	9.0248	0.9756
2.425	9.2449	8.9878	0.9722
2.430	9.2408	8.9781	0.9716
2.435	9.2405	8.9771	0.9715
2.440	9.2201	8.9581	0.9716
2.445	9.2103	8.9534	0.9721
2.450	9.2079	8.9498	0.9720
2.455	9.1920	8.9277	0.9712
2.460	9.1847	8.9198	0.9712
2.465	9.1702	8.9274	0.9735
2.470	9.1655	8.8909	0.9700
2.475	9.1589	8.8850	0.9701
2.480	9.1411	8.8763	0.9710
2.485	9.1368	8.8773	0.9716
2.490	9.1268	8.8548	0.9702
2.495	9.1114	8.8489	0.9712
2.500	9.1165	8.8384	0.9695
2.505	9.1050	8.8214	0.9689
2.510	9.0916	8.8146	0.9695
2.515	9.0843	8.8000	0.9687
2.520	9.0701	8.8005	0.9703
2.525	9.0506	8.7737	0.9694
2.530	9.0468	8.7706	0.9695
2.535	9.0533	8.7708	0.9688
2.540	9.0400	8.7459	0.9675
2.545	9.0338	8.7293	0.9663
2.550	9.0126	8.7197	0.9675
2.555	9.0202	8.7116	0.9658
2.560	8.9957	8.7043	0.9676
2.565	8.9853	8.6896	0.9671
2.570	8.9826	8.6775	0.9660
2.575	8.9745	8.6743	0.9665
2.580	8.9600	8.6500	0.9654
2.585	8.9403	8.6540	0.9680
2.590	8.9418	8.6261	0.9647
2.595	8.9226	8.6260	0.9668
2.600	8.9162	8.6009	0.9646
2.605	8.8990	8.5820	0.9644
2.610	8.9101	8.5904	0.9641
2.615	8.9066	8.5652	0.9617

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	8.8770	8.5602	0.9643
2.625	8.8678	8.5410	0.9631
2.630	8.8591	8.5445	0.9645
2.635	8.8720	8.5274	0.9612
2.640	8.8372	8.5103	0.9630
2.645	8.8392	8.5056	0.9623
2.650	8.8117	8.4921	0.9637
2.655	8.8277	8.4704	0.9595
2.660	8.7979	8.4704	0.9628
2.665	8.8130	8.4695	0.9610
2.670	8.8076	8.4439	0.9587
2.675	8.7808	8.4421	0.9614
2.680	8.7788	8.4309	0.9604
2.685	8.7554	8.4175	0.9614
2.690	8.7497	8.4033	0.9604
2.695	8.7362	8.3980	0.9613
2.700	8.7273	8.3954	0.9620
2.705	8.7249	8.3776	0.9602
2.710	8.7043	8.3827	0.9631
2.715	8.7044	8.3624	0.9607
2.720	8.6886	8.3570	0.9618
2.725	8.6752	8.3403	0.9614
2.730	8.6600	8.3250	0.9613
2.735	8.6469	8.3336	0.9638
2.740	8.6620	8.3058	0.9589
2.745	8.6337	8.3212	0.9638
2.750	8.6266	8.3044	0.9627
2.755	8.6108	8.2970	0.9636
2.760	8.6182	8.2962	0.9626
2.765	8.6039	8.2718	0.9614
2.770	8.5933	8.2842	0.9640
2.775	8.5863	8.2501	0.9609
2.780	8.5838	8.2493	0.9610
2.785	8.5691	8.2416	0.9618
2.790	8.5709	8.2252	0.9597
2.795	8.5451	8.2310	0.9632
2.800	8.5438	8.2152	0.9615
2.805	8.5394	8.2156	0.9621
2.810	8.5168	8.1883	0.9614
2.815	8.5118	8.1925	0.9625
2.820	8.5149	8.1774	0.9604
2.825	8.5051	8.1565	0.9590
2.830	8.4966	8.1650	0.9610
2.835	8.4972	8.1583	0.9601
2.840	8.4717	8.1486	0.9619
2.845	8.4621	8.1394	0.9619

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	8.4791	8.1155	0.9571
2.855	8.4606	8.1207	0.9598
2.860	8.4467	8.1024	0.9592
2.865	8.4270	8.0978	0.9609
2.870	8.4202	8.0858	0.9603
2.875	8.4242	8.0689	0.9578
2.880	8.4023	8.0757	0.9611
2.885	8.4002	8.0730	0.9610
2.890	8.3756	8.0687	0.9634
2.895	8.3852	8.0382	0.9586
2.900	8.3818	8.0381	0.9590
2.905	8.3653	8.0472	0.9620
2.910	8.3514	8.0370	0.9624
2.915	8.3504	8.0188	0.9603
2.920	8.3479	8.0139	0.9600
2.925	8.3572	7.9928	0.9564
2.930	8.3487	7.9860	0.9566
2.935	8.3278	7.9791	0.9581
2.940	8.3395	7.9768	0.9565
2.945	8.3206	7.9691	0.9578
2.950	8.3076	7.9532	0.9573
2.955	8.3386	7.9581	0.9544
2.960	8.2646	7.9397	0.9607
2.965	8.3101	7.9246	0.9536
2.970	8.2853	7.8986	0.9533
2.975	8.2852	7.8998	0.9535
2.980	8.2906	7.9023	0.9532
2.985	8.2712	7.8802	0.9527
2.990	8.2475	7.8872	0.9563
2.995	8.2600	7.8643	0.9521
3.000	8.2298	7.8628	0.9554
3.005	8.2432	7.8553	0.9529
3.010	8.2372	7.8215	0.9495
3.015	8.2314	7.8166	0.9496
3.020	8.1980	7.8340	0.9556
3.025	8.2086	7.8122	0.9517
3.030	8.2189	7.7979	0.9488
3.035	8.2167	7.8124	0.9508
3.040	8.1890	7.7928	0.9516
3.045	8.1851	7.7822	0.9508
3.050	8.1581	7.7512	0.9501
3.055	8.1645	7.7908	0.9542
3.060	8.1575	7.7771	0.9534
3.065	8.1817	7.7677	0.9494
3.070	8.1482	7.7634	0.9528
3.075	8.1547	7.7669	0.9524

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	8.1520	7.7613	0.9521
3.085	8.1487	7.7291	0.9485
3.090	8.1471	7.7382	0.9498
3.095	8.1473	7.7368	0.9496
3.100	8.1538	7.7248	0.9474
3.105	8.1357	7.6992	0.9463
3.110	8.1336	7.7092	0.9478
3.115	8.1300	7.6968	0.9467
3.120	8.1199	7.6962	0.9478
3.125	8.1141	7.6803	0.9465
3.130	8.1107	7.6640	0.9449
3.135	8.0895	7.6486	0.9455
3.140	8.1035	7.6490	0.9439
3.145	8.0858	7.6227	0.9427
3.150	8.0837	7.6226	0.9430
3.155	8.0732	7.6175	0.9436
3.160	8.0815	7.6053	0.9411
3.165	8.0534	7.5999	0.9437
3.170	8.0752	7.5807	0.9388
3.175	8.0365	7.5666	0.9415
3.180	8.0498	7.5699	0.9404
3.185	8.0244	7.5707	0.9435
3.190	8.0304	7.5558	0.9409
3.195	8.0056	7.5487	0.9429
3.200	7.9912	7.5371	0.9432
3.205	7.9943	7.5500	0.9444
3.210	7.9923	7.5301	0.9422
3.215	8.0002	7.5223	0.9403
3.220	8.0101	7.5366	0.9409
3.225	7.9636	7.5331	0.9459
3.230	7.9835	7.5031	0.9398
3.235	7.9699	7.5072	0.9419
3.240	7.9534	7.4862	0.9413
3.245	7.9626	7.4813	0.9396
3.250	7.9129	7.4933	0.9470
3.255	7.9277	7.4575	0.9407
3.260	7.9398	7.4639	0.9401
3.265	7.9267	7.4705	0.9424
3.270	7.9209	7.4123	0.9358
3.275	7.9106	7.4184	0.9378
3.280	7.9137	7.4293	0.9388
3.285	7.8648	7.3988	0.9408
3.290	7.8984	7.3778	0.9341
3.295	7.8782	7.3721	0.9358
3.300	7.8862	7.3794	0.9357
3.305	7.8717	7.3836	0.9380

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	7.8636	7.3610	0.9361
3.315	7.8445	7.3431	0.9361
3.320	7.8501	7.3541	0.9368
3.325	7.8579	7.3442	0.9346
3.330	7.8276	7.3353	0.9371
3.335	7.8155	7.3071	0.9349
3.340	7.8166	7.3148	0.9358
3.345	7.8053	7.2915	0.9342
3.350	7.7865	7.2864	0.9358
3.355	7.7979	7.2819	0.9338
3.360	7.7689	7.2811	0.9372
3.365	7.7706	7.2717	0.9358
3.370	7.7529	7.2623	0.9367
3.375	7.7669	7.2721	0.9363
3.380	7.7598	7.2610	0.9357
3.385	7.7459	7.2533	0.9364
3.390	7.7411	7.2379	0.9350
3.395	7.7379	7.2384	0.9354
3.400	7.7064	7.2309	0.9383
3.405	7.7104	7.2149	0.9357
3.410	7.7058	7.2249	0.9376
3.415	7.6914	7.2072	0.9370
3.420	7.7019	7.2004	0.9349
3.425	7.7028	7.1939	0.9339
3.430	7.7069	7.1933	0.9334
3.435	7.6853	7.1998	0.9368
3.440	7.6980	7.1694	0.9313
3.445	7.6764	7.1661	0.9335
3.450	7.6500	7.1495	0.9346
3.455	7.6665	7.1613	0.9341
3.460	7.6546	7.1482	0.9338
3.465	7.6550	7.1356	0.9322
3.470	7.6546	7.1225	0.9305
3.475	7.6536	7.1076	0.9287
3.480	7.6549	7.0826	0.9252
3.485	7.6620	7.0589	0.9213
3.490	7.6222	7.0777	0.9286
3.495	7.6104	7.0860	0.9311
3.500	7.6177	7.0548	0.9261
3.505	7.6308	7.0717	0.9267
3.510	7.6266	7.0248	0.9211
3.515	7.5884	7.0418	0.9280
3.520	7.5906	7.0064	0.9230
3.525	7.5252	7.0155	0.9323
3.530	7.5446	7.0308	0.9319
3.535	7.5267	7.0036	0.9305

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	7.5077	7.0130	0.9341
3.545	7.5144	7.0294	0.9355
3.550	7.5053	7.0292	0.9366
3.555	7.5078	7.0299	0.9363
3.560	7.5109	7.0406	0.9374
3.565	7.5252	7.0421	0.9358
3.570	7.4915	7.0443	0.9403
3.575	7.4737	7.1173	0.9523
3.580	7.5233	7.0810	0.9412
3.585	7.5852	7.0869	0.9343
3.590	7.6028	6.9998	0.9207
3.595	7.6170	6.9905	0.9177
3.600	7.5935	6.9961	0.9213
3.605	7.6213	6.9536	0.9124
3.610	7.6107	6.9389	0.9117
3.615	7.5710	6.9172	0.9136
3.620	7.5579	6.8834	0.9108
3.625	7.5999	6.8392	0.8999
3.630	7.4999	6.8299	0.9107
3.635	7.5017	6.8557	0.9139
3.640	7.5092	6.8688	0.9147
3.645	7.4925	6.8671	0.9165
3.650	7.4608	6.8560	0.9189
3.655	7.4544	6.8405	0.9177
3.660	7.4444	6.8630	0.9219
3.665	7.4502	6.8490	0.9193
3.670	7.4250	6.8511	0.9227
3.675	7.4407	6.8432	0.9197
3.680	7.4331	6.8491	0.9214
3.685	7.4261	6.8540	0.9230
3.690	7.4291	6.8525	0.9224
3.695	7.4235	6.8626	0.9244
3.700	7.4371	6.8548	0.9217
3.705	7.4501	6.8499	0.9194
3.710	7.4371	6.8380	0.9194
3.715	7.4600	6.8370	0.9165
3.720	7.4787	6.8160	0.9114
3.725	7.4555	6.8160	0.9142
3.730	7.4669	6.7860	0.9088
3.735	7.4524	6.7518	0.9060
3.740	7.4504	6.7385	0.9044
3.745	7.4288	6.7272	0.9056
3.750	7.3962	6.7156	0.9080
3.755	7.3701	6.7255	0.9125
3.760	7.3440	6.7439	0.9183
3.765	7.3261	6.7690	0.9240

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	7.3490	6.8097	0.9266
3.775	7.3837	6.8096	0.9223
3.780	7.3079	6.8277	0.9343
3.785	7.4169	6.8005	0.9169
3.790	7.4441	6.7440	0.9060
3.795	7.4332	6.7306	0.9055
3.800	7.4043	6.7488	0.9115
3.805	7.4176	6.7032	0.9037
3.810	7.3873	6.6872	0.9052
3.815	7.3827	6.6536	0.9012
3.820	7.3904	6.6847	0.9045
3.825	7.3691	6.6439	0.9016
3.830	7.3702	6.6575	0.9033
3.835	7.2501	6.4019	0.8830
3.840	7.3497	6.6515	0.9050
3.845	7.3352	6.6476	0.9063
3.850	7.3465	6.6382	0.9036
3.855	7.3267	6.6346	0.9055
3.860	7.3214	6.6193	0.9041
3.865	7.2966	6.6159	0.9067
3.870	7.3205	6.6213	0.9045
3.875	7.3014	6.6085	0.9051
3.880	7.3130	6.6042	0.9031
3.885	7.2924	6.5984	0.9048
3.890	7.2930	6.6047	0.9056
3.895	7.2977	6.5806	0.9017
3.900	7.2955	6.5712	0.9007
3.905	7.2655	6.5774	0.9053
3.910	7.2833	6.5677	0.9018
3.915	7.2725	6.5633	0.9025
3.920	7.2625	6.5637	0.9038
3.925	7.2607	6.5446	0.9014
3.930	7.2555	6.5415	0.9016
3.935	7.2599	6.5325	0.8998
3.940	7.2445	6.5305	0.9014
3.945	7.2404	6.5183	0.9003
3.950	7.2341	6.5088	0.8997
3.955	7.2246	6.5122	0.9014
3.960	7.2257	6.4986	0.8994
3.965	7.2268	6.4972	0.8990
3.970	7.2137	6.4967	0.9006
3.975	7.2045	6.4851	0.9001
3.980	7.2032	6.4801	0.8996
3.985	7.2065	6.4808	0.8993
3.990	7.2021	6.4727	0.8987
3.995	7.1912	6.4614	0.8985

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	7.1854	6.4562	0.8985
4.005	7.1841	6.4518	0.8981
4.010	7.1693	6.4436	0.8988
4.015	7.1731	6.4496	0.8991
4.020	7.1703	6.4320	0.8970
4.025	7.1817	6.4227	0.8943
4.030	7.1599	6.4201	0.8967
4.035	7.1465	6.4191	0.8982
4.040	7.1536	6.4113	0.8962
4.045	7.1568	6.3969	0.8938
4.050	7.1435	6.3924	0.8948
4.055	7.1386	6.3846	0.8944
4.060	7.1240	6.3791	0.8954
4.065	7.1152	6.3643	0.8945
4.070	7.1227	6.3781	0.8955
4.075	7.1065	6.3629	0.8954
4.080	7.0963	6.3712	0.8978
4.085	7.1058	6.3587	0.8949
4.090	7.1047	6.3543	0.8944
4.095	7.0785	6.3575	0.8981
4.100	7.0846	6.3544	0.8969
4.105	7.0879	6.3192	0.8915
4.110	7.0823	6.3321	0.8941
4.115	7.0728	6.3213	0.8937
4.120	7.0563	6.3141	0.8948
4.125	7.0774	6.3145	0.8922
4.130	7.0743	6.3205	0.8935
4.135	7.0553	6.2835	0.8906
4.140	7.0697	6.2741	0.8875
4.145	7.0887	6.2847	0.8866
4.150	7.0470	6.2751	0.8905
4.155	7.0346	6.2662	0.8908
4.160	7.0595	6.2676	0.8878
4.165	7.0358	6.2450	0.8876
4.170	7.0450	6.2349	0.8850
4.175	7.0685	6.2180	0.8797
4.180	7.0622	6.2341	0.8827
4.185	7.0530	6.2121	0.8808
4.190	7.0433	6.2120	0.8820
4.195	7.0423	6.2108	0.8819
4.200	7.0565	6.2159	0.8809
4.205	7.0503	6.2139	0.8814
4.210	7.0259	6.2063	0.8833
4.215	7.0506	6.2093	0.8807
4.220	7.0361	6.2153	0.8833
4.225	7.0710	6.2105	0.8783

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	7.0390	6.2040	0.8814
4.235	7.0284	6.1988	0.8820
4.240	7.0224	6.2218	0.8860
4.245	7.0390	6.2076	0.8819
4.250	7.0287	6.1932	0.8811
4.255	7.0287	6.2018	0.8824
4.260	7.0137	6.1925	0.8829
4.265	7.0333	6.1904	0.8802
4.270	7.0144	6.1805	0.8811
4.275	7.0075	6.1779	0.8816
4.280	7.0103	6.1819	0.8818
4.285	7.0041	6.1773	0.8819
4.290	6.9970	6.1744	0.8824
4.295	6.9992	6.1644	0.8807
4.300	6.9961	6.1714	0.8821
4.305	6.9907	6.1659	0.8820
4.310	6.9905	6.1585	0.8810
4.315	6.9865	6.1563	0.8812
4.320	6.9902	6.1516	0.8800
4.325	6.9854	6.1520	0.8807
4.330	6.9884	6.1470	0.8796
4.335	6.9813	6.1430	0.8799
4.340	6.9879	6.1366	0.8782
4.345	6.9764	6.1294	0.8786
4.350	6.9758	6.1243	0.8779
4.355	6.9713	6.1282	0.8791
4.360	6.9683	6.1164	0.8777
4.365	6.9575	6.1118	0.8785
4.370	6.9668	6.1100	0.8770
4.375	6.9486	6.1089	0.8792
4.380	6.9511	6.1024	0.8779
4.385	6.9275	6.0834	0.8781
4.390	6.9413	6.0746	0.8751
4.395	6.9269	6.0775	0.8774
4.400	6.9225	6.0786	0.8781
4.405	6.9062	6.0727	0.8793
4.410	6.9154	6.0723	0.8781
4.415	6.9120	6.0594	0.8767
4.420	6.8980	6.0737	0.8805
4.425	6.9005	6.0765	0.8806
4.430	6.8933	6.0526	0.8780
4.435	6.8808	6.0371	0.8774
4.440	6.8832	6.0436	0.8780
4.445	6.8606	6.0278	0.8786
4.450	6.8749	6.0072	0.8738
4.455	6.8443	6.0070	0.8777

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	6.8581	5.9833	0.8724
4.465	6.8715	5.9623	0.8677
4.470	6.8701	5.9737	0.8695
4.475	6.8875	5.9567	0.8649
4.480	6.8938	5.9650	0.8653
4.485	6.8687	5.9319	0.8636
4.490	6.8741	5.9453	0.8649
4.495	6.8775	5.9437	0.8642
4.500	6.8681	5.9411	0.8650
4.505	6.8624	5.9330	0.8646
4.510	6.8378	5.9285	0.8670
4.515	6.8402	5.9456	0.8692
4.520	6.8342	5.9380	0.8689
4.525	6.8476	5.9384	0.8672
4.530	6.8249	5.9227	0.8678
4.535	6.8193	5.9312	0.8698
4.540	6.8286	5.9161	0.8664
4.545	6.8112	5.9048	0.8669
4.550	6.8186	5.8906	0.8639
4.555	6.8073	5.8968	0.8662
4.560	6.8091	5.8953	0.8658
4.565	6.8020	5.9008	0.8675
4.570	6.7997	5.8836	0.8653
4.575	6.7999	5.8815	0.8649
4.580	6.7954	5.8863	0.8662
4.585	6.7889	5.8768	0.8656
4.590	6.7877	5.8667	0.8643
4.595	6.7786	5.8693	0.8659
4.600	6.7777	5.8590	0.8645
4.605	6.7699	5.8526	0.8645
4.610	6.7670	5.8499	0.8645
4.615	6.7619	5.8406	0.8638
4.620	6.7662	5.8346	0.8623
4.625	6.7569	5.8397	0.8643
4.630	6.7588	5.8268	0.8621
4.635	6.7507	5.8173	0.8617
4.640	6.7441	5.8169	0.8625
4.645	6.7443	5.8121	0.8618
4.650	6.7416	5.8085	0.8616
4.655	6.7404	5.7879	0.8587
4.660	6.7480	5.7996	0.8594
4.665	6.7387	5.7933	0.8597
4.670	6.7096	5.7832	0.8619
4.675	6.7147	5.7614	0.8580
4.680	6.7196	5.7685	0.8585
4.685	6.7216	5.7631	0.8574

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	6.7234	5.7723	0.8585
4.695	6.7188	5.7475	0.8554
4.700	6.7039	5.7621	0.8595
4.705	6.7105	5.7068	0.8504
4.710	6.6701	5.7390	0.8604
4.715	6.6896	5.7414	0.8583
4.720	6.6831	5.7239	0.8565
4.725	6.6918	5.7354	0.8571
4.730	6.6922	5.7281	0.8559
4.735	6.6977	5.7449	0.8577
4.740	6.6952	5.7315	0.8561
4.745	6.6786	5.7254	0.8573
4.750	6.6986	5.7197	0.8539
4.755	6.6910	5.7278	0.8560
4.760	6.6839	5.7124	0.8546
4.765	6.7121	5.7242	0.8528
4.770	6.6994	5.7241	0.8544
4.775	6.6635	5.6888	0.8537
4.780	6.6819	5.6950	0.8523
4.785	6.6725	5.7413	0.8605
4.790	6.6808	5.6770	0.8497
4.795	6.6693	5.6768	0.8512
4.800	6.6762	5.6849	0.8515
4.805	6.6876	5.6527	0.8452
4.810	6.6640	5.6856	0.8532
4.815	6.6869	5.6424	0.8438
4.820	6.6506	5.6639	0.8516
4.825	6.6743	5.6659	0.8489
4.830	6.6514	5.6586	0.8507
4.835	6.6489	5.6670	0.8523
4.840	6.6516	5.6645	0.8516
4.845	6.6553	5.6494	0.8489
4.850	6.6371	5.6500	0.8513
4.855	6.6526	5.6601	0.8508
4.860	6.6438	5.6485	0.8502
4.865	6.6676	5.6455	0.8467
4.870	6.6472	5.6375	0.8481
4.875	6.6449	5.6437	0.8493
4.880	6.6411	5.6508	0.8509
4.885	6.6396	5.6319	0.8482
4.890	6.6515	5.6467	0.8489
4.895	6.6345	5.6258	0.8480
4.900	6.6348	5.6346	0.8493
4.905	6.6364	5.6260	0.8478
4.910	6.6267	5.6198	0.8481
4.915	6.6249	5.6171	0.8479

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	6.6250	5.6123	0.8471
4.925	6.6242	5.6065	0.8464
4.930	6.6146	5.5998	0.8466
4.935	6.6153	5.5952	0.8458
4.940	6.6148	5.5912	0.8453
4.945	6.6099	5.5860	0.8451
4.950	6.6122	5.5831	0.8444
4.955	6.6138	5.5851	0.8445
4.960	6.6048	5.5741	0.8439
4.965	6.6130	5.5709	0.8424
4.970	6.6065	5.5742	0.8437
4.975	6.6132	5.5744	0.8429
4.980	6.5997	5.5680	0.8437
4.985	6.5962	5.5544	0.8421
4.990	6.5965	5.5629	0.8433
4.995	6.5792	5.5581	0.8448
5.000	6.5878	5.5530	0.8429
5.005	6.5776	5.5529	0.8442
5.010	6.5731	5.5454	0.8436
5.015	6.5698	5.5522	0.8451
5.020	6.5644	5.5344	0.8431
5.025	6.5450	5.5304	0.8450
5.030	6.5365	5.5201	0.8445
5.035	6.5383	5.5314	0.8460
5.040	6.5269	5.5155	0.8450
5.045	6.5308	5.5031	0.8426
5.050	6.5265	5.4806	0.8397
5.055	6.5390	5.4895	0.8395
5.060	6.5306	5.4787	0.8389
5.065	6.5401	5.4602	0.8349
5.070	6.5256	5.4765	0.8392
5.075	6.5257	5.4635	0.8372
5.080	6.5156	5.4526	0.8368
5.085	6.5129	5.4444	0.8359
5.090	6.5140	5.4570	0.8377
5.095	6.5145	5.4639	0.8387
5.100	6.5182	5.4512	0.8363
5.105	6.5005	5.4709	0.8416
5.110	6.4990	5.4392	0.8369
5.115	6.4925	5.4315	0.8366
5.120	6.4905	5.4286	0.8364
5.125	6.4895	5.4261	0.8361
5.130	6.4875	5.4125	0.8343
5.135	6.4752	5.4039	0.8346
5.140	6.4795	5.4123	0.8353
5.145	6.4730	5.3870	0.8322

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	6.4694	5.4028	0.8351
5.155	6.4698	5.3877	0.8327
5.160	6.4663	5.3826	0.8324
5.165	6.4616	5.3779	0.8323
5.170	6.4619	5.3670	0.8306
5.175	6.4533	5.3598	0.8306
5.180	6.4581	5.3582	0.8297
5.185	6.4532	5.3513	0.8292
5.190	6.4558	5.3492	0.8286
5.195	6.4573	5.3421	0.8273
5.200	6.4583	5.3430	0.8273
5.205	6.4605	5.3411	0.8267
5.210	6.4594	5.3357	0.8260
5.215	6.4608	5.3377	0.8262
5.220	6.4546	5.3301	0.8258
5.225	6.4478	5.3271	0.8262
5.230	6.4548	5.3218	0.8245
5.235	6.4449	5.3232	0.8260
5.240	6.4424	5.3180	0.8255
5.245	6.4441	5.3185	0.8253
5.250	6.4363	5.3208	0.8267
5.255	6.4242	5.3173	0.8277
5.260	6.4434	5.3095	0.8240
5.265	6.4455	5.3224	0.8257
5.270	6.4323	5.3087	0.8253
5.275	6.4234	5.2936	0.8241
5.280	6.4266	5.3031	0.8252
5.285	6.4323	5.2971	0.8235
5.290	6.4180	5.2888	0.8241
5.295	6.4382	5.2750	0.8193
5.300	6.4197	5.2807	0.8226
5.305	6.4266	5.2839	0.8222
5.310	6.3991	5.2723	0.8239
5.315	6.4197	5.2771	0.8220
5.320	6.4057	5.2602	0.8212
5.325	6.4132	5.2652	0.8210
5.330	6.4117	5.2610	0.8205
5.335	6.3864	5.2619	0.8239
5.340	6.3963	5.2679	0.8236
5.345	6.3853	5.2666	0.8248
5.350	6.4097	5.2522	0.8194
5.355	6.3900	5.2470	0.8211
5.360	6.3886	5.2188	0.8169
5.365	6.4070	5.2443	0.8185
5.370	6.3757	5.2182	0.8184
5.375	6.3941	5.2277	0.8176

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	6.3920	5.2351	0.8190
5.385	6.3764	5.2226	0.8191
5.390	6.3724	5.2304	0.8208
5.395	6.3872	5.2289	0.8187
5.400	6.3756	5.2219	0.8190
5.405	6.3707	5.2183	0.8191
5.410	6.3715	5.2168	0.8188
5.415	6.3662	5.1866	0.8147
5.420	6.3611	5.2140	0.8197
5.425	6.3619	5.2159	0.8199
5.430	6.3749	5.2033	0.8162
5.435	6.3701	5.1924	0.8151
5.440	6.3530	5.2007	0.8186
5.445	6.3716	5.1974	0.8157
5.450	6.3529	5.1891	0.8168
5.455	6.3644	5.1974	0.8166
5.460	6.3579	5.1820	0.8150
5.465	6.3670	5.2015	0.8169
5.470	6.3390	5.1794	0.8171
5.475	6.3514	5.1710	0.8142
5.480	6.3409	5.1777	0.8166
5.485	6.3496	5.1702	0.8142
5.490	6.3305	5.1763	0.8177
5.495	6.3482	5.1656	0.8137
5.500	6.3361	5.1723	0.8163
5.505	6.3413	5.1749	0.8161
5.510	6.3417	5.1640	0.8143
5.515	6.3436	5.1633	0.8139
5.520	6.3359	5.1563	0.8138
5.525	6.3424	5.1566	0.8130
5.530	6.3405	5.1519	0.8125
5.535	6.3436	5.1487	0.8116
5.540	6.3365	5.1499	0.8127
5.545	6.3298	5.1399	0.8120
5.550	6.3331	5.1391	0.8115
5.555	6.3390	5.1332	0.8098
5.560	6.3346	5.1329	0.8103
5.565	6.3357	5.1328	0.8101
5.570	6.3373	5.1332	0.8100
5.575	6.3382	5.1284	0.8091
5.580	6.3369	5.1375	0.8107
5.585	6.3339	5.1298	0.8099
5.590	6.3420	5.1239	0.8079
5.595	6.3364	5.1184	0.8078
5.600	6.3405	5.1216	0.8078
5.605	6.3466	5.1145	0.8059

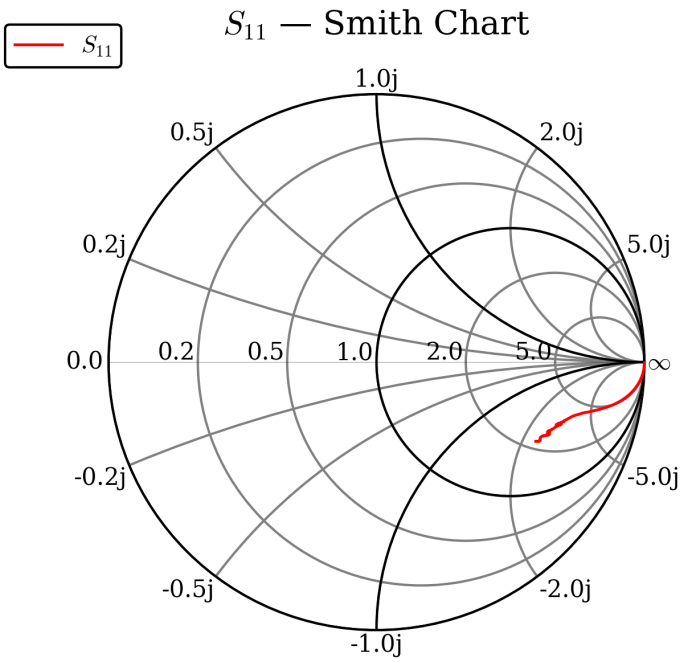
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	6.3377	5.1043	0.8054
5.615	6.3338	5.0882	0.8033
5.620	6.3326	5.0967	0.8048
5.625	6.3368	5.1000	0.8048
5.630	6.3252	5.0885	0.8045
5.635	6.3286	5.0811	0.8029
5.640	6.3298	5.0658	0.8003
5.645	6.3201	5.0645	0.8013
5.650	6.3235	5.0742	0.8024
5.655	6.3143	5.0652	0.8022
5.660	6.3118	5.0669	0.8028
5.665	6.3336	5.0642	0.7996
5.670	6.2948	5.0578	0.8035
5.675	6.3096	5.0429	0.7992
5.680	6.3033	5.0449	0.8004
5.685	6.3123	5.0339	0.7975
5.690	6.2991	5.0346	0.7993
5.695	6.2981	5.0511	0.8020
5.700	6.2967	5.0172	0.7968
5.705	6.2760	5.0108	0.7984
5.710	6.2980	5.0122	0.7958
5.715	6.2897	4.9959	0.7943
5.720	6.2842	5.0063	0.7966
5.725	6.2701	4.9998	0.7974
5.730	6.2705	4.9940	0.7964
5.735	6.2739	4.9897	0.7953
5.740	6.2712	4.9940	0.7963
5.745	6.2722	4.9908	0.7957
5.750	6.2591	4.9826	0.7961
5.755	6.2667	4.9812	0.7949
5.760	6.2515	4.9674	0.7946
5.765	6.2678	4.9661	0.7923
5.770	6.2613	4.9523	0.7909
5.775	6.2578	4.9622	0.7930
5.780	6.2461	4.9575	0.7937
5.785	6.2420	4.9550	0.7938
5.790	6.2333	4.9585	0.7955
5.795	6.2354	4.9595	0.7954
5.800	6.2264	4.9509	0.7951
5.805	6.2269	4.9428	0.7938
5.810	6.2238	4.9473	0.7949
5.815	6.2243	4.9457	0.7946
5.820	6.2226	4.9373	0.7934
5.825	6.2185	4.9335	0.7934
5.830	6.2092	4.9307	0.7941
5.835	6.2129	4.9273	0.7931

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	6.2074	4.9262	0.7936
5.845	6.2001	4.9268	0.7946
5.850	6.2056	4.9182	0.7925
5.855	6.2004	4.9115	0.7921
5.860	6.1952	4.9145	0.7933
5.865	6.1933	4.9163	0.7938
5.870	6.1955	4.9019	0.7912
5.875	6.1920	4.9010	0.7915
5.880	6.1895	4.8992	0.7915
5.885	6.1937	4.8890	0.7893
5.890	6.1750	4.8962	0.7929
5.895	6.1801	4.8880	0.7909
5.900	6.1794	4.8806	0.7898
5.905	6.1819	4.8922	0.7914
5.910	6.1772	4.8656	0.7877
5.915	6.1726	4.8709	0.7891
5.920	6.1690	4.8649	0.7886
5.925	6.1658	4.8547	0.7874
5.930	6.1568	4.8618	0.7897
5.935	6.1546	4.8534	0.7886
5.940	6.1542	4.8557	0.7890
5.945	6.1771	4.8437	0.7841
5.950	6.1438	4.8531	0.7899
5.955	6.1518	4.8585	0.7898
5.960	6.1468	4.8476	0.7886
5.965	6.1476	4.8550	0.7897
5.970	6.1431	4.8562	0.7905
5.975	6.1571	4.8239	0.7835
5.980	6.1750	4.8394	0.7837
5.985	6.1273	4.8313	0.7885
5.990	6.1296	4.8293	0.7879
5.995	6.1362	4.8212	0.7857
6.000	6.1477	4.8266	0.7851

1.2 S_{11} — Smith Chart



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0000	-0.0024	0.00
0.010	0.9999	-0.0049	-0.00
0.015	0.9998	-0.0074	-0.00
0.020	0.9996	-0.0101	-0.00
0.025	0.9995	-0.0124	-0.00
0.030	0.9991	-0.0150	-0.01
0.035	0.9990	-0.0176	-0.01
0.040	0.9986	-0.0201	-0.01
0.045	0.9983	-0.0225	-0.01
0.050	0.9979	-0.0251	-0.02
0.055	0.9974	-0.0274	-0.02
0.060	0.9970	-0.0300	-0.02
0.065	0.9965	-0.0324	-0.03
0.070	0.9959	-0.0347	-0.03
0.075	0.9954	-0.0370	-0.03
0.080	0.9947	-0.0394	-0.04
0.085	0.9940	-0.0419	-0.04
0.090	0.9934	-0.0442	-0.05
0.095	0.9925	-0.0465	-0.06
0.100	0.9920	-0.0488	-0.06
0.105	0.9910	-0.0511	-0.07
0.110	0.9904	-0.0533	-0.07
0.115	0.9895	-0.0557	-0.08

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9886	-0.0578	-0.08
0.125	0.9877	-0.0600	-0.09
0.130	0.9866	-0.0620	-0.10
0.135	0.9859	-0.0643	-0.11
0.140	0.9848	-0.0664	-0.11
0.145	0.9839	-0.0685	-0.12
0.150	0.9827	-0.0705	-0.13
0.155	0.9818	-0.0725	-0.14
0.160	0.9807	-0.0745	-0.14
0.165	0.9797	-0.0763	-0.15
0.170	0.9785	-0.0784	-0.16
0.175	0.9774	-0.0803	-0.17
0.180	0.9762	-0.0822	-0.18
0.185	0.9750	-0.0840	-0.19
0.190	0.9738	-0.0859	-0.20
0.195	0.9726	-0.0877	-0.21
0.200	0.9715	-0.0895	-0.21
0.205	0.9702	-0.0914	-0.22
0.210	0.9689	-0.0930	-0.23
0.215	0.9677	-0.0947	-0.24
0.220	0.9662	-0.0964	-0.26
0.225	0.9653	-0.0980	-0.26
0.230	0.9638	-0.0997	-0.27
0.235	0.9625	-0.1012	-0.28
0.240	0.9612	-0.1029	-0.29
0.245	0.9600	-0.1043	-0.30
0.250	0.9586	-0.1059	-0.31
0.255	0.9573	-0.1073	-0.32
0.260	0.9559	-0.1089	-0.34
0.265	0.9545	-0.1103	-0.35
0.270	0.9532	-0.1117	-0.36
0.275	0.9519	-0.1131	-0.37
0.280	0.9506	-0.1144	-0.38
0.285	0.9491	-0.1157	-0.39
0.290	0.9479	-0.1170	-0.40
0.295	0.9464	-0.1183	-0.41
0.300	0.9452	-0.1195	-0.42
0.305	0.9438	-0.1208	-0.43
0.310	0.9424	-0.1221	-0.44
0.315	0.9410	-0.1232	-0.45
0.320	0.9397	-0.1244	-0.47
0.325	0.9383	-0.1256	-0.48
0.330	0.9370	-0.1268	-0.49
0.335	0.9356	-0.1278	-0.50
0.340	0.9343	-0.1289	-0.51
0.345	0.9329	-0.1299	-0.52

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9315	-0.1309	-0.53
0.355	0.9303	-0.1319	-0.54
0.360	0.9289	-0.1329	-0.55
0.365	0.9277	-0.1338	-0.56
0.370	0.9262	-0.1348	-0.57
0.375	0.9248	-0.1357	-0.59
0.380	0.9236	-0.1368	-0.60
0.385	0.9222	-0.1376	-0.61
0.390	0.9209	-0.1386	-0.62
0.395	0.9197	-0.1394	-0.63
0.400	0.9185	-0.1402	-0.64
0.405	0.9172	-0.1411	-0.65
0.410	0.9159	-0.1419	-0.66
0.415	0.9146	-0.1427	-0.67
0.420	0.9133	-0.1434	-0.68
0.425	0.9121	-0.1442	-0.69
0.430	0.9108	-0.1448	-0.70
0.435	0.9097	-0.1455	-0.71
0.440	0.9085	-0.1463	-0.72
0.445	0.9073	-0.1469	-0.73
0.450	0.9061	-0.1476	-0.74
0.455	0.9050	-0.1483	-0.75
0.460	0.9038	-0.1489	-0.76
0.465	0.9026	-0.1496	-0.77
0.470	0.9013	-0.1502	-0.78
0.475	0.9003	-0.1509	-0.79
0.480	0.8990	-0.1514	-0.80
0.485	0.8980	-0.1519	-0.81
0.490	0.8968	-0.1525	-0.82
0.495	0.8957	-0.1531	-0.83
0.500	0.8947	-0.1537	-0.84
0.505	0.8936	-0.1541	-0.85
0.510	0.8925	-0.1546	-0.86
0.515	0.8914	-0.1552	-0.87
0.520	0.8905	-0.1558	-0.88
0.525	0.8894	-0.1561	-0.89
0.530	0.8884	-0.1566	-0.89
0.535	0.8874	-0.1571	-0.90
0.540	0.8864	-0.1575	-0.91
0.545	0.8854	-0.1580	-0.92
0.550	0.8843	-0.1586	-0.93
0.555	0.8833	-0.1591	-0.94
0.560	0.8823	-0.1593	-0.95
0.565	0.8813	-0.1598	-0.96
0.570	0.8803	-0.1602	-0.97
0.575	0.8794	-0.1607	-0.97

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8785	-0.1610	-0.98
0.585	0.8776	-0.1615	-0.99
0.590	0.8766	-0.1618	-1.00
0.595	0.8757	-0.1622	-1.01
0.600	0.8747	-0.1626	-1.02
0.605	0.8738	-0.1629	-1.02
0.610	0.8730	-0.1633	-1.03
0.615	0.8720	-0.1636	-1.04
0.620	0.8712	-0.1640	-1.05
0.625	0.8703	-0.1644	-1.05
0.630	0.8694	-0.1647	-1.06
0.635	0.8686	-0.1650	-1.07
0.640	0.8677	-0.1654	-1.08
0.645	0.8669	-0.1657	-1.08
0.650	0.8660	-0.1660	-1.09
0.655	0.8651	-0.1662	-1.10
0.660	0.8643	-0.1666	-1.11
0.665	0.8635	-0.1669	-1.12
0.670	0.8627	-0.1672	-1.12
0.675	0.8618	-0.1674	-1.13
0.680	0.8610	-0.1677	-1.14
0.685	0.8603	-0.1679	-1.14
0.690	0.8595	-0.1682	-1.15
0.695	0.8587	-0.1684	-1.16
0.700	0.8579	-0.1687	-1.17
0.705	0.8570	-0.1690	-1.17
0.710	0.8563	-0.1693	-1.18
0.715	0.8556	-0.1695	-1.19
0.720	0.8548	-0.1697	-1.20
0.725	0.8540	-0.1700	-1.20
0.730	0.8531	-0.1702	-1.21
0.735	0.8524	-0.1705	-1.22
0.740	0.8515	-0.1707	-1.22
0.745	0.8509	-0.1710	-1.23
0.750	0.8502	-0.1712	-1.24
0.755	0.8494	-0.1715	-1.24
0.760	0.8486	-0.1716	-1.25
0.765	0.8479	-0.1718	-1.26
0.770	0.8470	-0.1720	-1.27
0.775	0.8463	-0.1723	-1.27
0.780	0.8456	-0.1725	-1.28
0.785	0.8448	-0.1728	-1.29
0.790	0.8442	-0.1731	-1.29
0.795	0.8433	-0.1732	-1.30
0.800	0.8426	-0.1734	-1.31
0.805	0.8418	-0.1737	-1.31

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.8410	-0.1739	-1.32
0.815	0.8403	-0.1740	-1.33
0.820	0.8396	-0.1743	-1.34
0.825	0.8389	-0.1744	-1.34
0.830	0.8381	-0.1746	-1.35
0.835	0.8373	-0.1749	-1.36
0.840	0.8367	-0.1751	-1.36
0.845	0.8359	-0.1753	-1.37
0.850	0.8352	-0.1755	-1.38
0.855	0.8345	-0.1756	-1.38
0.860	0.8337	-0.1758	-1.39
0.865	0.8330	-0.1760	-1.40
0.870	0.8323	-0.1762	-1.40
0.875	0.8315	-0.1764	-1.41
0.880	0.8309	-0.1766	-1.42
0.885	0.8302	-0.1768	-1.42
0.890	0.8294	-0.1770	-1.43
0.895	0.8287	-0.1772	-1.44
0.900	0.8280	-0.1773	-1.45
0.905	0.8273	-0.1775	-1.45
0.910	0.8266	-0.1777	-1.46
0.915	0.8259	-0.1778	-1.46
0.920	0.8251	-0.1780	-1.47
0.925	0.8245	-0.1782	-1.48
0.930	0.8238	-0.1784	-1.48
0.935	0.8231	-0.1786	-1.49
0.940	0.8223	-0.1788	-1.50
0.945	0.8216	-0.1789	-1.51
0.950	0.8209	-0.1791	-1.51
0.955	0.8202	-0.1793	-1.52
0.960	0.8196	-0.1794	-1.52
0.965	0.8188	-0.1796	-1.53
0.970	0.8183	-0.1798	-1.54
0.975	0.8175	-0.1799	-1.54
0.980	0.8169	-0.1801	-1.55
0.985	0.8162	-0.1804	-1.56
0.990	0.8155	-0.1805	-1.56
0.995	0.8148	-0.1807	-1.57
1.000	0.8142	-0.1808	-1.58
1.005	0.8134	-0.1810	-1.58
1.010	0.8129	-0.1811	-1.59
1.015	0.8122	-0.1812	-1.60
1.020	0.8115	-0.1814	-1.60
1.025	0.8108	-0.1816	-1.61
1.030	0.8103	-0.1818	-1.61
1.035	0.8096	-0.1819	-1.62

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.040	0.8089	-0.1821	-1.63
1.045	0.8084	-0.1823	-1.63
1.050	0.8077	-0.1823	-1.64
1.055	0.8071	-0.1824	-1.64
1.060	0.8065	-0.1825	-1.65
1.065	0.8060	-0.1827	-1.66
1.070	0.8053	-0.1829	-1.66
1.075	0.8048	-0.1830	-1.67
1.080	0.8041	-0.1831	-1.67
1.085	0.8035	-0.1833	-1.68
1.090	0.8029	-0.1834	-1.69
1.095	0.8024	-0.1835	-1.69
1.100	0.8019	-0.1838	-1.70
1.105	0.8012	-0.1839	-1.70
1.110	0.8007	-0.1840	-1.71
1.115	0.8002	-0.1842	-1.71
1.120	0.7997	-0.1842	-1.72
1.125	0.7992	-0.1844	-1.72
1.130	0.7986	-0.1845	-1.73
1.135	0.7981	-0.1846	-1.73
1.140	0.7977	-0.1846	-1.74
1.145	0.7971	-0.1848	-1.74
1.150	0.7966	-0.1850	-1.75
1.155	0.7962	-0.1850	-1.75
1.160	0.7956	-0.1851	-1.76
1.165	0.7952	-0.1853	-1.76
1.170	0.7948	-0.1854	-1.77
1.175	0.7942	-0.1854	-1.77
1.180	0.7938	-0.1856	-1.77
1.185	0.7934	-0.1856	-1.78
1.190	0.7929	-0.1857	-1.78
1.195	0.7925	-0.1858	-1.79
1.200	0.7922	-0.1860	-1.79
1.205	0.7918	-0.1859	-1.80
1.210	0.7914	-0.1860	-1.80
1.215	0.7909	-0.1861	-1.80
1.220	0.7905	-0.1861	-1.81
1.225	0.7901	-0.1863	-1.81
1.230	0.7897	-0.1864	-1.82
1.235	0.7893	-0.1864	-1.82
1.240	0.7889	-0.1865	-1.82
1.245	0.7886	-0.1865	-1.83
1.250	0.7881	-0.1866	-1.83
1.255	0.7879	-0.1867	-1.83
1.260	0.7875	-0.1866	-1.84
1.265	0.7872	-0.1867	-1.84

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.7869	-0.1868	-1.84
1.275	0.7864	-0.1869	-1.85
1.280	0.7862	-0.1868	-1.85
1.285	0.7858	-0.1870	-1.85
1.290	0.7854	-0.1870	-1.86
1.295	0.7850	-0.1871	-1.86
1.300	0.7848	-0.1871	-1.86
1.305	0.7843	-0.1873	-1.87
1.310	0.7842	-0.1872	-1.87
1.315	0.7838	-0.1874	-1.87
1.320	0.7835	-0.1875	-1.88
1.325	0.7832	-0.1874	-1.88
1.330	0.7829	-0.1874	-1.88
1.335	0.7826	-0.1874	-1.89
1.340	0.7823	-0.1876	-1.89
1.345	0.7821	-0.1876	-1.89
1.350	0.7817	-0.1875	-1.90
1.355	0.7816	-0.1876	-1.90
1.360	0.7811	-0.1876	-1.90
1.365	0.7809	-0.1877	-1.90
1.370	0.7806	-0.1877	-1.91
1.375	0.7804	-0.1877	-1.91
1.380	0.7799	-0.1877	-1.91
1.385	0.7796	-0.1878	-1.92
1.390	0.7795	-0.1879	-1.92
1.395	0.7791	-0.1878	-1.92
1.400	0.7787	-0.1879	-1.93
1.405	0.7785	-0.1880	-1.93
1.410	0.7780	-0.1880	-1.93
1.415	0.7778	-0.1880	-1.94
1.420	0.7774	-0.1882	-1.94
1.425	0.7771	-0.1882	-1.94
1.430	0.7767	-0.1882	-1.95
1.435	0.7762	-0.1883	-1.95
1.440	0.7761	-0.1885	-1.95
1.445	0.7757	-0.1886	-1.96
1.450	0.7753	-0.1887	-1.96
1.455	0.7750	-0.1888	-1.96
1.460	0.7747	-0.1888	-1.97
1.465	0.7743	-0.1889	-1.97
1.470	0.7739	-0.1889	-1.98
1.475	0.7736	-0.1891	-1.98
1.480	0.7731	-0.1891	-1.98
1.485	0.7727	-0.1893	-1.99
1.490	0.7723	-0.1894	-1.99
1.495	0.7719	-0.1895	-1.99

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.500	0.7716	-0.1896	-2.00
1.505	0.7712	-0.1898	-2.00
1.510	0.7707	-0.1899	-2.01
1.515	0.7703	-0.1899	-2.01
1.520	0.7698	-0.1899	-2.02
1.525	0.7694	-0.1902	-2.02
1.530	0.7689	-0.1903	-2.02
1.535	0.7685	-0.1906	-2.03
1.540	0.7680	-0.1905	-2.03
1.545	0.7676	-0.1908	-2.04
1.550	0.7672	-0.1909	-2.04
1.555	0.7666	-0.1909	-2.05
1.560	0.7660	-0.1911	-2.05
1.565	0.7654	-0.1913	-2.06
1.570	0.7651	-0.1915	-2.06
1.575	0.7646	-0.1916	-2.07
1.580	0.7642	-0.1918	-2.07
1.585	0.7635	-0.1919	-2.08
1.590	0.7631	-0.1920	-2.08
1.595	0.7626	-0.1922	-2.09
1.600	0.7621	-0.1924	-2.09
1.605	0.7615	-0.1925	-2.10
1.610	0.7612	-0.1926	-2.10
1.615	0.7607	-0.1928	-2.11
1.620	0.7601	-0.1928	-2.11
1.625	0.7598	-0.1932	-2.11
1.630	0.7593	-0.1932	-2.12
1.635	0.7586	-0.1934	-2.13
1.640	0.7581	-0.1936	-2.13
1.645	0.7576	-0.1936	-2.14
1.650	0.7572	-0.1939	-2.14
1.655	0.7567	-0.1940	-2.15
1.660	0.7560	-0.1943	-2.15
1.665	0.7555	-0.1944	-2.16
1.670	0.7550	-0.1946	-2.16
1.675	0.7545	-0.1949	-2.17
1.680	0.7540	-0.1949	-2.17
1.685	0.7534	-0.1951	-2.18
1.690	0.7530	-0.1953	-2.18
1.695	0.7524	-0.1954	-2.19
1.700	0.7519	-0.1955	-2.19
1.705	0.7514	-0.1955	-2.20
1.710	0.7510	-0.1958	-2.20
1.715	0.7504	-0.1959	-2.21
1.720	0.7499	-0.1961	-2.21
1.725	0.7494	-0.1963	-2.22

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.730	0.7488	-0.1964	-2.22
1.735	0.7483	-0.1966	-2.23
1.740	0.7478	-0.1967	-2.23
1.745	0.7473	-0.1971	-2.24
1.750	0.7468	-0.1970	-2.24
1.755	0.7463	-0.1973	-2.25
1.760	0.7460	-0.1974	-2.25
1.765	0.7456	-0.1977	-2.26
1.770	0.7450	-0.1978	-2.26
1.775	0.7445	-0.1977	-2.27
1.780	0.7443	-0.1979	-2.27
1.785	0.7438	-0.1977	-2.27
1.790	0.7434	-0.1981	-2.28
1.795	0.7430	-0.1982	-2.28
1.800	0.7426	-0.1982	-2.29
1.805	0.7424	-0.1984	-2.29
1.810	0.7418	-0.1986	-2.29
1.815	0.7416	-0.1987	-2.30
1.820	0.7412	-0.1988	-2.30
1.825	0.7409	-0.1990	-2.30
1.830	0.7406	-0.1990	-2.31
1.835	0.7402	-0.1991	-2.31
1.840	0.7399	-0.1993	-2.31
1.845	0.7396	-0.1994	-2.32
1.850	0.7393	-0.1994	-2.32
1.855	0.7389	-0.1993	-2.32
1.860	0.7386	-0.1995	-2.33
1.865	0.7383	-0.1994	-2.33
1.870	0.7379	-0.1995	-2.33
1.875	0.7376	-0.1996	-2.34
1.880	0.7374	-0.1998	-2.34
1.885	0.7371	-0.1999	-2.34
1.890	0.7369	-0.2000	-2.34
1.895	0.7367	-0.2000	-2.35
1.900	0.7366	-0.2001	-2.35
1.905	0.7364	-0.2001	-2.35
1.910	0.7361	-0.2001	-2.35
1.915	0.7359	-0.2002	-2.35
1.920	0.7357	-0.2003	-2.36
1.925	0.7355	-0.2004	-2.36
1.930	0.7354	-0.2006	-2.36
1.935	0.7353	-0.2006	-2.36
1.940	0.7351	-0.2006	-2.36
1.945	0.7349	-0.2007	-2.36
1.950	0.7348	-0.2009	-2.36
1.955	0.7347	-0.2009	-2.36

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.960	0.7346	-0.2010	-2.37
1.965	0.7345	-0.2011	-2.37
1.970	0.7343	-0.2011	-2.37
1.975	0.7342	-0.2010	-2.37
1.980	0.7340	-0.2011	-2.37
1.985	0.7340	-0.2012	-2.37
1.990	0.7338	-0.2011	-2.37
1.995	0.7337	-0.2012	-2.38
2.000	0.7335	-0.2012	-2.38
2.005	0.7334	-0.2012	-2.38
2.010	0.7334	-0.2014	-2.38
2.015	0.7332	-0.2014	-2.38
2.020	0.7331	-0.2013	-2.38
2.025	0.7330	-0.2014	-2.38
2.030	0.7330	-0.2016	-2.38
2.035	0.7328	-0.2017	-2.38
2.040	0.7326	-0.2017	-2.38
2.045	0.7325	-0.2018	-2.39
2.050	0.7325	-0.2018	-2.39
2.055	0.7325	-0.2018	-2.39
2.060	0.7322	-0.2019	-2.39
2.065	0.7322	-0.2021	-2.39
2.070	0.7320	-0.2020	-2.39
2.075	0.7320	-0.2021	-2.39
2.080	0.7316	-0.2021	-2.39
2.085	0.7316	-0.2022	-2.40
2.090	0.7313	-0.2023	-2.40
2.095	0.7313	-0.2023	-2.40
2.100	0.7311	-0.2024	-2.40
2.105	0.7309	-0.2026	-2.40
2.110	0.7307	-0.2026	-2.40
2.115	0.7305	-0.2026	-2.41
2.120	0.7302	-0.2029	-2.41
2.125	0.7301	-0.2030	-2.41
2.130	0.7298	-0.2031	-2.41
2.135	0.7296	-0.2032	-2.41
2.140	0.7293	-0.2034	-2.42
2.145	0.7291	-0.2035	-2.42
2.150	0.7288	-0.2037	-2.42
2.155	0.7286	-0.2038	-2.42
2.160	0.7282	-0.2040	-2.43
2.165	0.7280	-0.2041	-2.43
2.170	0.7278	-0.2042	-2.43
2.175	0.7274	-0.2044	-2.43
2.180	0.7271	-0.2046	-2.44
2.185	0.7268	-0.2046	-2.44

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.7264	-0.2048	-2.44
2.195	0.7262	-0.2051	-2.45
2.200	0.7259	-0.2053	-2.45
2.205	0.7255	-0.2054	-2.45
2.210	0.7253	-0.2057	-2.45
2.215	0.7248	-0.2058	-2.46
2.220	0.7245	-0.2060	-2.46
2.225	0.7240	-0.2062	-2.47
2.230	0.7238	-0.2065	-2.47
2.235	0.7232	-0.2066	-2.47
2.240	0.7229	-0.2068	-2.48
2.245	0.7224	-0.2070	-2.48
2.250	0.7221	-0.2072	-2.48
2.255	0.7218	-0.2074	-2.49
2.260	0.7213	-0.2076	-2.49
2.265	0.7208	-0.2078	-2.50
2.270	0.7206	-0.2080	-2.50
2.275	0.7201	-0.2083	-2.50
2.280	0.7196	-0.2086	-2.51
2.285	0.7193	-0.2088	-2.51
2.290	0.7189	-0.2089	-2.51
2.295	0.7186	-0.2092	-2.52
2.300	0.7181	-0.2093	-2.52
2.305	0.7175	-0.2097	-2.53
2.310	0.7171	-0.2097	-2.53
2.315	0.7167	-0.2101	-2.53
2.320	0.7163	-0.2104	-2.54
2.325	0.7158	-0.2106	-2.54
2.330	0.7152	-0.2107	-2.55
2.335	0.7149	-0.2111	-2.55
2.340	0.7145	-0.2114	-2.56
2.345	0.7140	-0.2117	-2.56
2.350	0.7137	-0.2120	-2.56
2.355	0.7133	-0.2122	-2.57
2.360	0.7128	-0.2124	-2.57
2.365	0.7123	-0.2125	-2.58
2.370	0.7121	-0.2128	-2.58
2.375	0.7115	-0.2131	-2.58
2.380	0.7111	-0.2131	-2.59
2.385	0.7106	-0.2133	-2.59
2.390	0.7103	-0.2137	-2.59
2.395	0.7099	-0.2140	-2.60
2.400	0.7094	-0.2143	-2.60
2.405	0.7091	-0.2144	-2.61
2.410	0.7088	-0.2146	-2.61
2.415	0.7083	-0.2149	-2.61

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.420	0.7079	-0.2151	-2.62
2.425	0.7077	-0.2153	-2.62
2.430	0.7073	-0.2155	-2.62
2.435	0.7069	-0.2156	-2.63
2.440	0.7066	-0.2157	-2.63
2.445	0.7063	-0.2160	-2.63
2.450	0.7060	-0.2161	-2.64
2.455	0.7056	-0.2162	-2.64
2.460	0.7054	-0.2164	-2.64
2.465	0.7050	-0.2164	-2.64
2.470	0.7047	-0.2167	-2.65
2.475	0.7045	-0.2168	-2.65
2.480	0.7041	-0.2168	-2.65
2.485	0.7039	-0.2171	-2.66
2.490	0.7036	-0.2171	-2.66
2.495	0.7033	-0.2173	-2.66
2.500	0.7031	-0.2174	-2.66
2.505	0.7028	-0.2176	-2.67
2.510	0.7026	-0.2177	-2.67
2.515	0.7023	-0.2179	-2.67
2.520	0.7019	-0.2180	-2.67
2.525	0.7017	-0.2181	-2.68
2.530	0.7015	-0.2182	-2.68
2.535	0.7012	-0.2184	-2.68
2.540	0.7010	-0.2185	-2.68
2.545	0.7008	-0.2187	-2.69
2.550	0.7006	-0.2187	-2.69
2.555	0.7004	-0.2189	-2.69
2.560	0.7002	-0.2188	-2.69
2.565	0.7000	-0.2187	-2.69
2.570	0.6999	-0.2187	-2.69
2.575	0.6998	-0.2186	-2.70
2.580	0.6999	-0.2185	-2.70
2.585	0.7001	-0.2185	-2.69
2.590	0.7002	-0.2185	-2.69
2.595	0.7003	-0.2187	-2.69
2.600	0.7004	-0.2190	-2.69
2.605	0.7004	-0.2190	-2.69
2.610	0.7004	-0.2192	-2.69
2.615	0.7005	-0.2195	-2.69
2.620	0.7005	-0.2195	-2.69
2.625	0.7006	-0.2197	-2.68
2.630	0.7006	-0.2198	-2.68
2.635	0.7005	-0.2200	-2.68
2.640	0.7006	-0.2201	-2.68
2.645	0.7005	-0.2201	-2.68

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	0.7006	-0.2201	-2.68
2.655	0.7006	-0.2205	-2.68
2.660	0.7005	-0.2204	-2.68
2.665	0.7004	-0.2205	-2.68
2.670	0.7003	-0.2207	-2.68
2.675	0.7001	-0.2207	-2.68
2.680	0.7002	-0.2209	-2.68
2.685	0.7001	-0.2209	-2.68
2.690	0.7001	-0.2211	-2.68
2.695	0.7002	-0.2212	-2.68
2.700	0.7001	-0.2213	-2.68
2.705	0.7002	-0.2214	-2.68
2.710	0.7003	-0.2216	-2.68
2.715	0.7003	-0.2215	-2.68
2.720	0.7003	-0.2216	-2.68
2.725	0.7001	-0.2216	-2.68
2.730	0.7002	-0.2217	-2.68
2.735	0.7002	-0.2215	-2.68
2.740	0.7001	-0.2215	-2.68
2.745	0.7001	-0.2216	-2.68
2.750	0.7000	-0.2216	-2.68
2.755	0.6998	-0.2215	-2.69
2.760	0.6998	-0.2216	-2.69
2.765	0.6996	-0.2216	-2.69
2.770	0.6997	-0.2215	-2.69
2.775	0.6994	-0.2216	-2.69
2.780	0.6992	-0.2215	-2.69
2.785	0.6992	-0.2216	-2.69
2.790	0.6991	-0.2216	-2.69
2.795	0.6989	-0.2215	-2.70
2.800	0.6987	-0.2216	-2.70
2.805	0.6985	-0.2217	-2.70
2.810	0.6984	-0.2217	-2.70
2.815	0.6982	-0.2217	-2.70
2.820	0.6981	-0.2219	-2.70
2.825	0.6980	-0.2220	-2.70
2.830	0.6977	-0.2220	-2.71
2.835	0.6974	-0.2222	-2.71
2.840	0.6973	-0.2222	-2.71
2.845	0.6972	-0.2224	-2.71
2.850	0.6969	-0.2226	-2.71
2.855	0.6969	-0.2226	-2.71
2.860	0.6966	-0.2228	-2.72
2.865	0.6964	-0.2228	-2.72
2.870	0.6964	-0.2229	-2.72
2.875	0.6964	-0.2231	-2.72

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.880	0.6962	-0.2228	-2.72
2.885	0.6961	-0.2227	-2.72
2.890	0.6959	-0.2229	-2.73
2.895	0.6957	-0.2228	-2.73
2.900	0.6953	-0.2228	-2.73
2.905	0.6949	-0.2228	-2.74
2.910	0.6948	-0.2226	-2.74
2.915	0.6945	-0.2227	-2.74
2.920	0.6941	-0.2228	-2.75
2.925	0.6937	-0.2230	-2.75
2.930	0.6932	-0.2231	-2.75
2.935	0.6930	-0.2232	-2.76
2.940	0.6926	-0.2236	-2.76
2.945	0.6923	-0.2238	-2.76
2.950	0.6920	-0.2239	-2.76
2.955	0.6918	-0.2242	-2.77
2.960	0.6916	-0.2244	-2.77
2.965	0.6913	-0.2246	-2.77
2.970	0.6912	-0.2248	-2.77
2.975	0.6910	-0.2251	-2.77
2.980	0.6909	-0.2255	-2.77
2.985	0.6910	-0.2255	-2.77
2.990	0.6909	-0.2256	-2.77
2.995	0.6909	-0.2258	-2.77
3.000	0.6908	-0.2258	-2.77
3.005	0.6899	-0.2257	-2.78
3.010	0.6899	-0.2261	-2.78
3.015	0.6899	-0.2263	-2.78
3.020	0.6899	-0.2264	-2.78
3.025	0.6900	-0.2269	-2.78
3.030	0.6902	-0.2269	-2.78
3.035	0.6903	-0.2272	-2.77
3.040	0.6905	-0.2269	-2.77
3.045	0.6906	-0.2270	-2.77
3.050	0.6908	-0.2266	-2.77
3.055	0.6909	-0.2266	-2.77
3.060	0.6908	-0.2264	-2.77
3.065	0.6907	-0.2261	-2.77
3.070	0.6907	-0.2260	-2.77
3.075	0.6904	-0.2256	-2.78
3.080	0.6903	-0.2256	-2.78
3.085	0.6901	-0.2254	-2.78
3.090	0.6897	-0.2252	-2.79
3.095	0.6893	-0.2251	-2.79
3.100	0.6891	-0.2252	-2.79
3.105	0.6889	-0.2249	-2.80

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	0.6885	-0.2251	-2.80
3.115	0.6881	-0.2250	-2.81
3.120	0.6878	-0.2253	-2.81
3.125	0.6875	-0.2255	-2.81
3.130	0.6874	-0.2259	-2.81
3.135	0.6873	-0.2260	-2.81
3.140	0.6871	-0.2264	-2.81
3.145	0.6871	-0.2268	-2.81
3.150	0.6871	-0.2270	-2.81
3.155	0.6871	-0.2274	-2.81
3.160	0.6872	-0.2277	-2.81
3.165	0.6873	-0.2279	-2.80
3.170	0.6877	-0.2282	-2.80
3.175	0.6879	-0.2281	-2.80
3.180	0.6879	-0.2287	-2.79
3.185	0.6882	-0.2287	-2.79
3.190	0.6886	-0.2285	-2.79
3.195	0.6888	-0.2284	-2.79
3.200	0.6888	-0.2282	-2.79
3.205	0.6890	-0.2282	-2.78
3.210	0.6892	-0.2280	-2.78
3.215	0.6891	-0.2278	-2.78
3.220	0.6890	-0.2275	-2.79
3.225	0.6889	-0.2271	-2.79
3.230	0.6885	-0.2271	-2.79
3.235	0.6884	-0.2274	-2.79
3.240	0.6883	-0.2276	-2.79
3.245	0.6881	-0.2275	-2.80
3.250	0.6881	-0.2277	-2.80
3.255	0.6880	-0.2282	-2.79
3.260	0.6881	-0.2285	-2.79
3.265	0.6881	-0.2287	-2.79
3.270	0.6883	-0.2287	-2.79
3.275	0.6883	-0.2290	-2.79
3.280	0.6881	-0.2291	-2.79
3.285	0.6885	-0.2295	-2.78
3.290	0.6886	-0.2296	-2.78
3.295	0.6886	-0.2300	-2.78
3.300	0.6887	-0.2298	-2.78
3.305	0.6886	-0.2301	-2.78
3.310	0.6888	-0.2304	-2.78
3.315	0.6890	-0.2303	-2.78
3.320	0.6892	-0.2306	-2.77
3.325	0.6892	-0.2310	-2.77
3.330	0.6894	-0.2309	-2.77
3.335	0.6893	-0.2311	-2.77

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	0.6896	-0.2312	-2.77
3.345	0.6897	-0.2313	-2.76
3.350	0.6897	-0.2313	-2.76
3.355	0.6899	-0.2313	-2.76
3.360	0.6899	-0.2313	-2.76
3.365	0.6900	-0.2314	-2.76
3.370	0.6901	-0.2314	-2.76
3.375	0.6901	-0.2313	-2.76
3.380	0.6901	-0.2313	-2.76
3.385	0.6900	-0.2313	-2.76
3.390	0.6900	-0.2312	-2.76
3.395	0.6899	-0.2311	-2.76
3.400	0.6898	-0.2310	-2.76
3.405	0.6898	-0.2309	-2.76
3.410	0.6896	-0.2307	-2.77
3.415	0.6894	-0.2308	-2.77
3.420	0.6891	-0.2308	-2.77
3.425	0.6891	-0.2307	-2.77
3.430	0.6887	-0.2309	-2.78
3.435	0.6885	-0.2306	-2.78
3.440	0.6882	-0.2308	-2.78
3.445	0.6878	-0.2308	-2.79
3.450	0.6874	-0.2308	-2.79
3.455	0.6870	-0.2310	-2.80
3.460	0.6867	-0.2311	-2.80
3.465	0.6862	-0.2313	-2.80
3.470	0.6860	-0.2316	-2.80
3.475	0.6857	-0.2316	-2.81
3.480	0.6854	-0.2321	-2.81
3.485	0.6851	-0.2321	-2.81
3.490	0.6853	-0.2331	-2.81
3.495	0.6847	-0.2330	-2.81
3.500	0.6847	-0.2334	-2.81
3.505	0.6848	-0.2339	-2.81
3.510	0.6847	-0.2342	-2.81
3.515	0.6850	-0.2341	-2.81
3.520	0.6853	-0.2346	-2.80
3.525	0.6854	-0.2347	-2.80
3.530	0.6857	-0.2347	-2.80
3.535	0.6860	-0.2341	-2.79
3.540	0.6863	-0.2340	-2.79
3.545	0.6869	-0.2335	-2.79
3.550	0.6864	-0.2330	-2.79
3.555	0.6862	-0.2321	-2.80
3.560	0.6861	-0.2313	-2.81
3.565	0.6854	-0.2304	-2.82

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.570	0.6843	-0.2299	-2.83
3.575	0.6824	-0.2283	-2.86
3.580	0.6823	-0.2283	-2.86
3.585	0.6802	-0.2279	-2.88
3.590	0.6792	-0.2287	-2.89
3.595	0.6774	-0.2294	-2.91
3.600	0.6765	-0.2305	-2.92
3.605	0.6757	-0.2318	-2.92
3.610	0.6755	-0.2330	-2.92
3.615	0.6756	-0.2341	-2.91
3.620	0.6761	-0.2352	-2.90
3.625	0.6770	-0.2361	-2.89
3.630	0.6773	-0.2360	-2.89
3.635	0.6780	-0.2360	-2.88
3.640	0.6783	-0.2359	-2.88
3.645	0.6785	-0.2357	-2.87
3.650	0.6791	-0.2351	-2.87
3.655	0.6791	-0.2348	-2.87
3.660	0.6790	-0.2345	-2.87
3.665	0.6788	-0.2343	-2.88
3.670	0.6788	-0.2337	-2.88
3.675	0.6785	-0.2336	-2.88
3.680	0.6785	-0.2331	-2.88
3.685	0.6784	-0.2328	-2.89
3.690	0.6779	-0.2326	-2.89
3.695	0.6776	-0.2322	-2.90
3.700	0.6773	-0.2320	-2.90
3.705	0.6768	-0.2318	-2.91
3.710	0.6761	-0.2316	-2.92
3.715	0.6755	-0.2315	-2.93
3.720	0.6748	-0.2314	-2.93
3.725	0.6739	-0.2314	-2.94
3.730	0.6728	-0.2319	-2.95
3.735	0.6720	-0.2323	-2.96
3.740	0.6713	-0.2333	-2.97
3.745	0.6709	-0.2342	-2.97
3.750	0.6706	-0.2355	-2.97
3.755	0.6707	-0.2367	-2.96
3.760	0.6711	-0.2378	-2.95
3.765	0.6719	-0.2388	-2.94
3.770	0.6734	-0.2393	-2.92
3.775	0.6746	-0.2394	-2.90
3.780	0.6764	-0.2388	-2.89
3.785	0.6766	-0.2376	-2.89
3.790	0.6768	-0.2361	-2.89
3.795	0.6764	-0.2353	-2.90

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	0.6758	-0.2344	-2.91
3.805	0.6752	-0.2339	-2.92
3.810	0.6745	-0.2336	-2.93
3.815	0.6735	-0.2339	-2.94
3.820	0.6725	-0.2340	-2.95
3.825	0.6723	-0.2342	-2.95
3.830	0.6716	-0.2348	-2.96
3.835	0.6711	-0.2365	-2.96
3.840	0.6711	-0.2369	-2.95
3.845	0.6713	-0.2370	-2.95
3.850	0.6710	-0.2375	-2.95
3.855	0.6712	-0.2376	-2.95
3.860	0.6710	-0.2377	-2.95
3.865	0.6708	-0.2378	-2.95
3.870	0.6708	-0.2380	-2.95
3.875	0.6708	-0.2383	-2.95
3.880	0.6707	-0.2384	-2.95
3.885	0.6706	-0.2387	-2.95
3.890	0.6706	-0.2390	-2.95
3.895	0.6704	-0.2389	-2.95
3.900	0.6702	-0.2396	-2.95
3.905	0.6700	-0.2396	-2.96
3.910	0.6698	-0.2398	-2.96
3.915	0.6696	-0.2400	-2.96
3.920	0.6696	-0.2405	-2.96
3.925	0.6693	-0.2407	-2.96
3.930	0.6692	-0.2408	-2.96
3.935	0.6691	-0.2411	-2.96
3.940	0.6687	-0.2414	-2.96
3.945	0.6685	-0.2415	-2.97
3.950	0.6682	-0.2420	-2.97
3.955	0.6679	-0.2422	-2.97
3.960	0.6677	-0.2425	-2.97
3.965	0.6674	-0.2427	-2.97
3.970	0.6671	-0.2430	-2.98
3.975	0.6669	-0.2431	-2.98
3.980	0.6665	-0.2434	-2.98
3.985	0.6661	-0.2438	-2.98
3.990	0.6659	-0.2440	-2.99
3.995	0.6654	-0.2443	-2.99
4.000	0.6651	-0.2447	-2.99
4.005	0.6646	-0.2451	-2.99
4.010	0.6643	-0.2453	-3.00
4.015	0.6638	-0.2456	-3.00
4.020	0.6635	-0.2459	-3.00
4.025	0.6633	-0.2462	-3.01

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	0.6626	-0.2466	-3.01
4.035	0.6623	-0.2467	-3.01
4.040	0.6620	-0.2470	-3.02
4.045	0.6615	-0.2473	-3.02
4.050	0.6609	-0.2475	-3.03
4.055	0.6603	-0.2478	-3.03
4.060	0.6601	-0.2481	-3.03
4.065	0.6598	-0.2484	-3.04
4.070	0.6590	-0.2487	-3.04
4.075	0.6587	-0.2486	-3.05
4.080	0.6583	-0.2490	-3.05
4.085	0.6576	-0.2494	-3.06
4.090	0.6570	-0.2496	-3.06
4.095	0.6566	-0.2496	-3.07
4.100	0.6563	-0.2500	-3.07
4.105	0.6553	-0.2504	-3.08
4.110	0.6553	-0.2508	-3.08
4.115	0.6548	-0.2511	-3.08
4.120	0.6543	-0.2512	-3.09
4.125	0.6539	-0.2518	-3.09
4.130	0.6530	-0.2524	-3.10
4.135	0.6528	-0.2528	-3.10
4.140	0.6518	-0.2530	-3.11
4.145	0.6509	-0.2534	-3.12
4.150	0.6503	-0.2538	-3.12
4.155	0.6499	-0.2539	-3.13
4.160	0.6490	-0.2544	-3.13
4.165	0.6485	-0.2543	-3.14
4.170	0.6479	-0.2548	-3.15
4.175	0.6473	-0.2552	-3.15
4.180	0.6468	-0.2552	-3.16
4.185	0.6463	-0.2550	-3.16
4.190	0.6457	-0.2553	-3.17
4.195	0.6452	-0.2554	-3.17
4.200	0.6448	-0.2556	-3.18
4.205	0.6445	-0.2559	-3.18
4.210	0.6441	-0.2561	-3.18
4.215	0.6437	-0.2563	-3.19
4.220	0.6432	-0.2562	-3.19
4.225	0.6429	-0.2569	-3.19
4.230	0.6426	-0.2566	-3.20
4.235	0.6423	-0.2568	-3.20
4.240	0.6420	-0.2569	-3.20
4.245	0.6416	-0.2573	-3.21
4.250	0.6415	-0.2573	-3.21
4.255	0.6414	-0.2574	-3.21

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.260	0.6409	-0.2576	-3.21
4.265	0.6409	-0.2578	-3.21
4.270	0.6408	-0.2580	-3.21
4.275	0.6406	-0.2581	-3.22
4.280	0.6403	-0.2584	-3.22
4.285	0.6402	-0.2585	-3.22
4.290	0.6399	-0.2584	-3.22
4.295	0.6399	-0.2587	-3.22
4.300	0.6397	-0.2588	-3.22
4.305	0.6396	-0.2589	-3.22
4.310	0.6394	-0.2589	-3.22
4.315	0.6393	-0.2590	-3.23
4.320	0.6391	-0.2591	-3.23
4.325	0.6388	-0.2592	-3.23
4.330	0.6387	-0.2594	-3.23
4.335	0.6386	-0.2594	-3.23
4.340	0.6383	-0.2597	-3.23
4.345	0.6382	-0.2598	-3.23
4.350	0.6382	-0.2601	-3.23
4.355	0.6379	-0.2603	-3.24
4.360	0.6379	-0.2606	-3.24
4.365	0.6378	-0.2609	-3.23
4.370	0.6378	-0.2610	-3.23
4.375	0.6379	-0.2614	-3.23
4.380	0.6381	-0.2615	-3.23
4.385	0.6383	-0.2618	-3.22
4.390	0.6384	-0.2620	-3.22
4.395	0.6385	-0.2622	-3.22
4.400	0.6387	-0.2624	-3.22
4.405	0.6390	-0.2625	-3.21
4.410	0.6391	-0.2626	-3.21
4.415	0.6392	-0.2625	-3.21
4.420	0.6393	-0.2625	-3.21
4.425	0.6395	-0.2621	-3.21
4.430	0.6397	-0.2623	-3.21
4.435	0.6397	-0.2624	-3.21
4.440	0.6397	-0.2623	-3.21
4.445	0.6397	-0.2621	-3.21
4.450	0.6398	-0.2623	-3.20
4.455	0.6402	-0.2622	-3.20
4.460	0.6401	-0.2620	-3.20
4.465	0.6403	-0.2623	-3.20
4.470	0.6405	-0.2620	-3.20
4.475	0.6408	-0.2624	-3.19
4.480	0.6411	-0.2624	-3.19
4.485	0.6416	-0.2628	-3.18

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.490	0.6421	-0.2630	-3.17
4.495	0.6423	-0.2634	-3.17
4.500	0.6427	-0.2637	-3.16
4.505	0.6430	-0.2640	-3.16
4.510	0.6435	-0.2643	-3.15
4.515	0.6439	-0.2645	-3.15
4.520	0.6444	-0.2645	-3.14
4.525	0.6446	-0.2645	-3.14
4.530	0.6450	-0.2646	-3.13
4.535	0.6452	-0.2644	-3.13
4.540	0.6454	-0.2645	-3.13
4.545	0.6457	-0.2644	-3.13
4.550	0.6459	-0.2645	-3.12
4.555	0.6459	-0.2646	-3.12
4.560	0.6461	-0.2646	-3.12
4.565	0.6461	-0.2646	-3.12
4.570	0.6463	-0.2647	-3.12
4.575	0.6464	-0.2646	-3.12
4.580	0.6464	-0.2645	-3.12
4.585	0.6466	-0.2646	-3.12
4.590	0.6466	-0.2646	-3.11
4.595	0.6466	-0.2646	-3.11
4.600	0.6467	-0.2646	-3.11
4.605	0.6465	-0.2647	-3.12
4.610	0.6466	-0.2646	-3.12
4.615	0.6465	-0.2647	-3.12
4.620	0.6462	-0.2647	-3.12
4.625	0.6462	-0.2647	-3.12
4.630	0.6462	-0.2649	-3.12
4.635	0.6460	-0.2649	-3.12
4.640	0.6458	-0.2651	-3.12
4.645	0.6457	-0.2648	-3.12
4.650	0.6458	-0.2651	-3.12
4.655	0.6457	-0.2651	-3.12
4.660	0.6457	-0.2654	-3.12
4.665	0.6457	-0.2653	-3.12
4.670	0.6457	-0.2652	-3.12
4.675	0.6458	-0.2652	-3.12
4.680	0.6455	-0.2653	-3.12
4.685	0.6455	-0.2654	-3.12
4.690	0.6456	-0.2657	-3.12
4.695	0.6457	-0.2654	-3.12
4.700	0.6456	-0.2653	-3.12
4.705	0.6457	-0.2661	-3.12
4.710	0.6456	-0.2654	-3.12
4.715	0.6454	-0.2654	-3.12

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.720	0.6453	-0.2652	-3.13
4.725	0.6452	-0.2655	-3.13
4.730	0.6451	-0.2651	-3.13
4.735	0.6446	-0.2650	-3.14
4.740	0.6446	-0.2649	-3.14
4.745	0.6440	-0.2650	-3.14
4.750	0.6438	-0.2650	-3.14
4.755	0.6436	-0.2652	-3.15
4.760	0.6429	-0.2653	-3.15
4.765	0.6425	-0.2657	-3.16
4.770	0.6421	-0.2658	-3.16
4.775	0.6418	-0.2655	-3.17
4.780	0.6412	-0.2659	-3.17
4.785	0.6412	-0.2662	-3.17
4.790	0.6410	-0.2660	-3.17
4.795	0.6407	-0.2664	-3.17
4.800	0.6403	-0.2665	-3.18
4.805	0.6406	-0.2665	-3.18
4.810	0.6403	-0.2660	-3.18
4.815	0.6406	-0.2661	-3.18
4.820	0.6401	-0.2656	-3.19
4.825	0.6400	-0.2656	-3.19
4.830	0.6398	-0.2654	-3.19
4.835	0.6398	-0.2649	-3.19
4.840	0.6398	-0.2645	-3.19
4.845	0.6399	-0.2642	-3.19
4.850	0.6398	-0.2642	-3.20
4.855	0.6398	-0.2638	-3.20
4.860	0.6396	-0.2636	-3.20
4.865	0.6395	-0.2638	-3.20
4.870	0.6395	-0.2638	-3.20
4.875	0.6396	-0.2636	-3.20
4.880	0.6394	-0.2635	-3.20
4.885	0.6394	-0.2636	-3.20
4.890	0.6394	-0.2637	-3.20
4.895	0.6396	-0.2638	-3.20
4.900	0.6396	-0.2638	-3.20
4.905	0.6396	-0.2640	-3.20
4.910	0.6397	-0.2641	-3.20
4.915	0.6398	-0.2643	-3.19
4.920	0.6399	-0.2645	-3.19
4.925	0.6400	-0.2647	-3.19
4.930	0.6400	-0.2649	-3.19
4.935	0.6401	-0.2652	-3.19
4.940	0.6402	-0.2654	-3.19
4.945	0.6402	-0.2657	-3.18

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	0.6403	-0.2660	-3.18
4.955	0.6402	-0.2663	-3.18
4.960	0.6403	-0.2664	-3.18
4.965	0.6403	-0.2669	-3.18
4.970	0.6402	-0.2670	-3.18
4.975	0.6401	-0.2672	-3.18
4.980	0.6402	-0.2677	-3.17
4.985	0.6403	-0.2679	-3.17
4.990	0.6402	-0.2682	-3.17
4.995	0.6403	-0.2686	-3.17
5.000	0.6401	-0.2690	-3.17
5.005	0.6400	-0.2693	-3.17
5.010	0.6398	-0.2695	-3.17
5.015	0.6398	-0.2699	-3.17
5.020	0.6398	-0.2699	-3.17
5.025	0.6396	-0.2702	-3.17
5.030	0.6395	-0.2703	-3.17
5.035	0.6394	-0.2700	-3.17
5.040	0.6395	-0.2700	-3.17
5.045	0.6397	-0.2698	-3.17
5.050	0.6398	-0.2698	-3.17
5.055	0.6399	-0.2696	-3.17
5.060	0.6403	-0.2695	-3.16
5.065	0.6408	-0.2696	-3.16
5.070	0.6409	-0.2691	-3.16
5.075	0.6414	-0.2692	-3.15
5.080	0.6417	-0.2689	-3.15
5.085	0.6420	-0.2687	-3.15
5.090	0.6422	-0.2685	-3.15
5.095	0.6423	-0.2683	-3.15
5.100	0.6428	-0.2686	-3.14
5.105	0.6430	-0.2683	-3.14
5.110	0.6433	-0.2685	-3.13
5.115	0.6434	-0.2685	-3.13
5.120	0.6438	-0.2686	-3.13
5.125	0.6439	-0.2688	-3.13
5.130	0.6443	-0.2689	-3.12
5.135	0.6445	-0.2689	-3.12
5.140	0.6445	-0.2691	-3.12
5.145	0.6449	-0.2693	-3.11
5.150	0.6449	-0.2693	-3.11
5.155	0.6449	-0.2695	-3.11
5.160	0.6451	-0.2697	-3.11
5.165	0.6451	-0.2698	-3.11
5.170	0.6452	-0.2700	-3.11
5.175	0.6452	-0.2700	-3.11

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	0.6451	-0.2701	-3.11
5.185	0.6452	-0.2703	-3.10
5.190	0.6451	-0.2704	-3.11
5.195	0.6450	-0.2705	-3.11
5.200	0.6449	-0.2706	-3.11
5.205	0.6448	-0.2708	-3.11
5.210	0.6446	-0.2710	-3.11
5.215	0.6443	-0.2711	-3.11
5.220	0.6442	-0.2712	-3.11
5.225	0.6441	-0.2713	-3.11
5.230	0.6438	-0.2713	-3.11
5.235	0.6437	-0.2715	-3.12
5.240	0.6433	-0.2715	-3.12
5.245	0.6431	-0.2714	-3.12
5.250	0.6427	-0.2715	-3.13
5.255	0.6424	-0.2714	-3.13
5.260	0.6421	-0.2714	-3.13
5.265	0.6416	-0.2716	-3.14
5.270	0.6412	-0.2716	-3.14
5.275	0.6410	-0.2715	-3.15
5.280	0.6404	-0.2716	-3.15
5.285	0.6399	-0.2716	-3.16
5.290	0.6396	-0.2716	-3.16
5.295	0.6391	-0.2719	-3.17
5.300	0.6388	-0.2719	-3.17
5.305	0.6383	-0.2719	-3.18
5.310	0.6379	-0.2721	-3.18
5.315	0.6373	-0.2726	-3.18
5.320	0.6370	-0.2724	-3.19
5.325	0.6368	-0.2725	-3.19
5.330	0.6364	-0.2728	-3.19
5.335	0.6360	-0.2725	-3.20
5.340	0.6355	-0.2732	-3.20
5.345	0.6357	-0.2732	-3.20
5.350	0.6350	-0.2731	-3.21
5.355	0.6346	-0.2732	-3.21
5.360	0.6344	-0.2732	-3.21
5.365	0.6338	-0.2729	-3.22
5.370	0.6336	-0.2728	-3.22
5.375	0.6332	-0.2732	-3.23
5.380	0.6324	-0.2728	-3.24
5.385	0.6317	-0.2728	-3.25
5.390	0.6314	-0.2727	-3.25
5.395	0.6307	-0.2729	-3.26
5.400	0.6302	-0.2730	-3.26
5.405	0.6296	-0.2731	-3.27

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.410	0.6291	-0.2734	-3.27
5.415	0.6292	-0.2735	-3.27
5.420	0.6280	-0.2735	-3.29
5.425	0.6274	-0.2739	-3.29
5.430	0.6269	-0.2740	-3.30
5.435	0.6267	-0.2743	-3.30
5.440	0.6262	-0.2744	-3.30
5.445	0.6256	-0.2745	-3.31
5.450	0.6253	-0.2747	-3.31
5.455	0.6248	-0.2749	-3.32
5.460	0.6248	-0.2750	-3.32
5.465	0.6239	-0.2751	-3.33
5.470	0.6240	-0.2751	-3.33
5.475	0.6236	-0.2750	-3.33
5.480	0.6232	-0.2750	-3.33
5.485	0.6225	-0.2752	-3.34
5.490	0.6222	-0.2752	-3.35
5.495	0.6216	-0.2752	-3.35
5.500	0.6211	-0.2750	-3.36
5.505	0.6204	-0.2752	-3.37
5.510	0.6201	-0.2755	-3.37
5.515	0.6194	-0.2755	-3.38
5.520	0.6190	-0.2758	-3.38
5.525	0.6187	-0.2759	-3.38
5.530	0.6184	-0.2764	-3.38
5.535	0.6180	-0.2764	-3.39
5.540	0.6177	-0.2767	-3.39
5.545	0.6173	-0.2769	-3.39
5.550	0.6171	-0.2771	-3.40
5.555	0.6168	-0.2774	-3.40
5.560	0.6165	-0.2775	-3.40
5.565	0.6163	-0.2775	-3.40
5.570	0.6159	-0.2777	-3.41
5.575	0.6157	-0.2779	-3.41
5.580	0.6152	-0.2780	-3.41
5.585	0.6149	-0.2782	-3.42
5.590	0.6145	-0.2784	-3.42
5.595	0.6142	-0.2787	-3.42
5.600	0.6137	-0.2789	-3.43
5.605	0.6132	-0.2793	-3.43
5.610	0.6129	-0.2797	-3.43
5.615	0.6128	-0.2800	-3.43
5.620	0.6121	-0.2804	-3.44
5.625	0.6119	-0.2813	-3.43
5.630	0.6116	-0.2813	-3.44
5.635	0.6116	-0.2819	-3.43

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.640	0.6115	-0.2824	-3.43
5.645	0.6113	-0.2829	-3.43
5.650	0.6113	-0.2834	-3.43
5.655	0.6110	-0.2837	-3.43
5.660	0.6109	-0.2841	-3.43
5.665	0.6108	-0.2845	-3.43
5.670	0.6108	-0.2851	-3.43
5.675	0.6107	-0.2854	-3.43
5.680	0.6104	-0.2860	-3.43
5.685	0.6105	-0.2862	-3.42
5.690	0.6103	-0.2865	-3.42
5.695	0.6100	-0.2870	-3.42
5.700	0.6102	-0.2877	-3.42
5.705	0.6103	-0.2880	-3.42
5.710	0.6101	-0.2883	-3.42
5.715	0.6101	-0.2888	-3.41
5.720	0.6099	-0.2890	-3.42
5.725	0.6101	-0.2895	-3.41
5.730	0.6100	-0.2898	-3.41
5.735	0.6101	-0.2903	-3.41
5.740	0.6100	-0.2908	-3.40
5.745	0.6105	-0.2912	-3.40
5.750	0.6102	-0.2915	-3.40
5.755	0.6101	-0.2917	-3.40
5.760	0.6102	-0.2920	-3.39
5.765	0.6103	-0.2924	-3.39
5.770	0.6103	-0.2926	-3.39
5.775	0.6102	-0.2927	-3.39
5.780	0.6102	-0.2931	-3.39
5.785	0.6100	-0.2933	-3.39
5.790	0.6100	-0.2934	-3.39
5.795	0.6099	-0.2935	-3.39
5.800	0.6099	-0.2936	-3.39
5.805	0.6098	-0.2938	-3.39
5.810	0.6098	-0.2939	-3.39
5.815	0.6096	-0.2941	-3.39
5.820	0.6095	-0.2942	-3.39
5.825	0.6093	-0.2943	-3.39
5.830	0.6092	-0.2943	-3.39
5.835	0.6090	-0.2945	-3.40
5.840	0.6087	-0.2944	-3.40
5.845	0.6086	-0.2944	-3.40
5.850	0.6083	-0.2946	-3.40
5.855	0.6081	-0.2947	-3.40
5.860	0.6076	-0.2947	-3.41
5.865	0.6073	-0.2944	-3.41

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	0.6070	-0.2948	-3.42
5.875	0.6066	-0.2949	-3.42
5.880	0.6062	-0.2949	-3.43
5.885	0.6058	-0.2949	-3.43
5.890	0.6055	-0.2952	-3.43
5.895	0.6051	-0.2953	-3.44
5.900	0.6046	-0.2951	-3.44
5.905	0.6041	-0.2955	-3.45
5.910	0.6040	-0.2957	-3.45
5.915	0.6037	-0.2958	-3.45
5.920	0.6034	-0.2958	-3.45
5.925	0.6033	-0.2957	-3.45
5.930	0.6027	-0.2962	-3.46
5.935	0.6022	-0.2959	-3.47
5.940	0.6018	-0.2959	-3.47
5.945	0.6016	-0.2961	-3.47
5.950	0.6011	-0.2953	-3.48
5.955	0.6003	-0.2952	-3.49
5.960	0.6000	-0.2953	-3.49
5.965	0.5990	-0.2951	-3.51
5.970	0.5984	-0.2952	-3.51
5.975	0.5976	-0.2952	-3.52
5.980	0.5975	-0.2953	-3.52
5.985	0.5970	-0.2954	-3.53
5.990	0.5958	-0.2953	-3.54
5.995	0.5953	-0.2958	-3.55
6.000	0.5944	-0.2958	-3.56

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.